



A truncated Gaussian filter for data assimilation with inequality constraints: Application to the hydrostatic stability condition in ocean models

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ABSTRACT

In many data assimilation problems, the state variables are subjected to inequality constraints. These constraints often contain valuable information that must be taken into account in the estimation process. However, with linear estimation methods (like the Kalman filter), there is no way to incorporate optimally that kind of additional information. In this study, it is shown that an optimal filter dealing with inequality constraints can be formulated under the assumption that the probability distributions are truncated Gaussian distributions. The statistical tools needed to implement this truncated Gaussian filter are described. It is also shown how the filter can be adapted to work in a reduced dimension space, and how it can be simplified following several additional hypotheses. As an application, the truncated Gaussian assumption is shown to be adequate to deal with the condition of hydrostatic stability in ocean analyses. First, a detailed evaluation of the method is made using a one-dimensional z -coordinate model of the mixed layer: particular attention is paid to the parameterization of the probability distribution, the accuracy of the estimation and the sensitivity to the observation system. In a second step, the method is applied to a three-dimensional hybrid coordinate ocean model (HYCOM) of the Bay of Biscay (at a $1/15^\circ$ resolution), to show that it is efficient enough to be applied to real size problems. These examples also demonstrate that the algorithm can deal with the hydrostatic stability condition in isopycnic coordinates as well as in z -coordinates.

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1. Introduction

Linear estimation methods (like the Kalman filter) are either defined as the linear estimator minimizing the variance of the estimation error or based on the stronger assumption that the error probability distributions are Gaussian. This last assumption is indeed a sufficient condition for the error variance minimizing estimator to be linear and equal to the maximum probability estimator. Problems can occur if such linear methods are used to estimate a vector of variables subjected to inequality constraints. In such cases, the Gaussian assumption is not valid, and nothing prevents the best linear estimation from violating the inequality constraints. The information contained in the inequality constraints is not taken into account in the estimation process.

In many data assimilation problems, however, the state variables are subjected to inequality constraints that must be taken into account in the assimilation process because: (i) they can be an important source of information about the system, (ii) they are often a prerequisite for a valid model initial condition. For in-

stance, in ocean systems, gravitational stability means that the Brunt-Väisälä frequency $N^2 = -\frac{g}{\rho} \frac{\partial \rho}{\partial z}$ (where ρ is the potential density referenced to the local in situ pressure) remains positive. This condition often contains valuable information that can improve the accuracy of the estimation. The reason is that in the ocean there is most often a top layer (the mixed layer) that is marginally stable, so that the ocean state is usually close to a number of hydrostatic stability inequality constraints. Moreover, in isopycnic coordinate ocean models (representing the ocean state as a stack of layers of prescribed density, e.g. Bleck, 2002), the hydrostatic stability condition is expressed in terms of a constraint of non-negativity of the layer thicknesses. Estimating an ocean state with non-negative layer thicknesses is obviously a prerequisite for initializing a model forecast. Ocean models are also subjected to other kinds of inequality constraints: non-negativity of tracer concentration, sea-water temperature above freezing point, ... For these reasons, several solutions have been proposed in the literature for including inequality constraints in existing linear data assimilation schemes.

First, inequality constraints can be introduced in variational data assimilation methods by adding non-quadratic terms (i.e. with a nonlinear gradient) in the cost function to be minimized (e.g. Fujii et al., 2005). However, this amounts to changing the

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nature of the problem, replacing the strong constraints by weak constraints. Moreover, this solution requires using a variational algorithm to determine the minimum of a non-quadratic cost function.

A second way of dealing with inequality constraints is to make a nonlinear change of variables (anamorphosis) that transforms the constrained variables x to unconstrained variables x' (Bertino et al., 2002). For instance, for a non-negative variable $x \geq 0$ (like a concentration or a layer thickness), one possible solution is to use a logarithmic transformation (e.g. $x' = \log_{10} x$) and assume a Gaussian distribution in the logarithmic space. The problem with this method is that it is not easy to find the change of variables that leads to a reasonable assumption. If we assume, for example, that x' is distributed like $\mathcal{N}(0, 1)$, the mean $x' = 0$ corresponds to a unitary distance ($x = 1$) with respect to the constraint ($x \geq 0$), the left tail of the Gaussian corresponds to probability tending to zero close to the constraint (typically $x' < -3$ or $x < 10^{-3}$), and the right tail of the Gaussian corresponds to probability tending to zero far from the constraint (typically $x' > 3$ or $x > 10^3$). The resulting assumption for x is therefore zero probability close to the constraint ($x \sim 0$), and significant probability for very large values ($x \sim 10^3$). We will see that such features are not very appropriate for the application considered in this paper.

A third solution proposed in the literature for including inequality constraints in linear schemes is to apply the linear scheme as it is and to introduce an adjustment operator to restore the constraints whenever needed. In their assimilation system, Brankart et al. (2003) introduced an adjustment operator (see Section 5) to restore non-negative layer thicknesses in an isopycnic ocean model (MICOM) after the observational update. More recently, in a hybrid coordinate ocean model (HYCOM), Thacker (2007) proposed using the correlations contained in the background error covariance matrix to move the ocean state from the linear solution to an adjusted solution with non-negative layer thicknesses. His solution is similar to that proposed by Simon and Simon (2005) who solved a quadratic programming problem to compute the adjusted solution (see Section 2.6.1). However, the difficulty with these adjustment operations is that they are based on heuristic considerations, and are only used to compensate for problems resulting from the linearity assumption.

Apart from these transformed linear schemes, there is also the possibility of using more general nonlinear estimation methods like the particle filters (e.g. Van Leeuwen, 2003; Losa et al., 2003; Vossepoel and Van Leeuwen, 2007). Since there is no hypothesis about the shape of prior probability distributions (approximated by an ensemble of model states), the particle filter can potentially deal with any kind of problem and, among other things, solves the problem of dealing with inequality constraints. However, describing a general distribution with sufficient accuracy requires using large ensembles, which involves numerical costs that can become prohibitive for practical ocean data assimilation problems.

In this paper, we choose to develop an optimal nonlinear scheme like the particle filter, with the difference that one prior assumption is made about the shape of the probability distributions: we assume that they are truncated Gaussian distributions (i.e. basically Gaussian distributions truncated by the inequality constraints). Consequently, the resulting estimation will be nonlinear in the input data and the maximum probability estimator will become different from the error variance minimizing estimator. A side effect of these developments is that we will be able to interpret the solution proposed by Simon and Simon (2005) or Thacker (2007) as an approximation of the optimal scheme (in the maximum probability estimator version). In this way, this study can also be seen as an attempt to provide a more precise theoretical basis for the adjustment operations proposed in these

studies. These theoretical aspects are presented in Section 2 of this paper.

With regard to applications of the method, we will seek to determine whether it is able to deal adequately with the condition of hydrostatic stability in ocean analyses, using twin experiments (i) with a one-dimensional z-coordinate model of the ocean mixed layer (Sections 3 and 4) and (ii) with a three-dimensional hybrid coordinate model of the Bay of Biscay (Section 5). With the one-dimensional model, we will be able to describe in detail the parameterization of the probability distributions (Section 3) and the observational update of truncated Gaussian distributions (Section 4), while, with the three-dimensional model, only synthetic results will be presented (Section 5).

2. Methodology

The purpose of this section is to show that an optimal filter dealing with inequality constraints can be formulated, under the assumption that the error probability distributions are truncated Gaussian. For that purpose, we first give the definition and main properties of truncated Gaussian probability distributions (Section 2.1). Then we present the truncated Gaussian filter (TG filter), with the basic hypothesis sustaining it (Section 2.2). In the two following (Sections 2.3 and 2.4), we describe the statistical tools that are needed to effectively implement the filter. And finally, we show how the filter can work in a reduced dimension space (Section 2.5), or can be approximated using additional assumptions (Section 2.6).

2.1. Truncated Gaussian probability distributions

The truncated normal probability distribution can be defined from the normal distribution as follows (Tallis, 1965; Robert, 1995). If a random vector $\mathbf{x}$ is Gaussian with probability distribution $p(\mathbf{x}) = \mathcal{N}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}})$, where $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{C}}$ are the expected value and covariance of $\mathbf{x}$, then, the posterior probability distribution of $\mathbf{x}$ with the condition that a set of linear inequality constraints $I: \mathbf{Ax} \leq \mathbf{b}$ are satisfied is called a *truncated Gaussian probability distribution*. It is entirely described by the parameters $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{C}}$ of the original Gaussian process together with the inequality constraints I , and can be noted:

$$p(\mathbf{x}|I) = \mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$$

$\tilde{\mathbf{x}}$ and $\tilde{\mathbf{C}}$ are sometimes called the location vector and the scale matrix of the truncated Gaussian distribution. The truncated Gaussian probability density function (pdf) is proportional to the Gaussian pdf for $\mathbf{Ax} \leq \mathbf{b}$ and equal to zero for $\mathbf{Ax} > \mathbf{b}$:

$$\mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I) = \begin{cases} \frac{1}{\alpha} \mathcal{N}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}) & \mathbf{Ax} \leq \mathbf{b} \\ 0 & \mathbf{Ax} > \mathbf{b} \end{cases} \quad (1)$$

The normalizing factor α can be determined by integrating the Gaussian pdf over the region defined by the inequality constraints I :

$$\alpha = \int_{\mathbf{Ax} \leq \mathbf{b}} \mathcal{N}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}) d\mathbf{x} \quad (2)$$

The expected value $\bar{\mathbf{x}}$ and covariance $\mathbf{C}$ of the truncated Gaussian distribution $\mathcal{TN}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$ can be determined by integrating the Gaussian distribution over the region defined by the inequality constraints I :

$$\bar{\mathbf{x}} = \frac{1}{\alpha} \int_{\mathbf{Ax} \leq \mathbf{b}} \mathbf{x} \mathcal{N}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}) d\mathbf{x} \quad (3)$$

$$\mathbf{C} = \frac{1}{\alpha} \int_{\mathbf{Ax} \leq \mathbf{b}} (\mathbf{x} - \bar{\mathbf{x}}) \mathcal{N}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}) (\mathbf{x} - \bar{\mathbf{x}})^T d\mathbf{x} \quad (4)$$

If $\mathbf{x}$ is a random vector with truncated Gaussian pdf $\mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$, the expected value $\bar{\mathbf{x}}$ also satisfies the inequality constraints: $\mathbf{A}\bar{\mathbf{x}} \leq \mathbf{b}$. (This is a direct consequence of the convexity of the domain defined by a set of linear inequality constraints.) Integrals (3) and (4) can be computed numerically by Monte Carlo integration, using the sampling algorithm described in Section 2.3. The mode $\hat{\mathbf{x}}$ of the truncated Gaussian distribution $\mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$ can be determined from its parameters $\tilde{\mathbf{x}}, \tilde{\mathbf{C}}$ and I by solving the following quadratic programming problem:

$$\min_{\mathbf{x}} (\tilde{\mathbf{x}} - \mathbf{x})^T \tilde{\mathbf{C}}^{-1} (\tilde{\mathbf{x}} - \mathbf{x}) \quad \text{such that} \quad \mathbf{A}\mathbf{x} \leq \mathbf{b} \quad (5)$$

which can be rewritten

$$\min_{\mathbf{x}} (\tilde{\mathbf{x}} \tilde{\mathbf{C}}^{-1} \tilde{\mathbf{x}}^T - 2\tilde{\mathbf{x}} \tilde{\mathbf{C}}^{-1} \mathbf{x}^T) \quad \text{such that} \quad \mathbf{A}\mathbf{x} \leq \mathbf{b} \quad (6)$$

If $\mathbf{x}$ is a random vector with probability distribution $\mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$, the maximum probability estimator of $\mathbf{x}$ is $\hat{\mathbf{x}}$, and the error variance minimizing estimator of $\mathbf{x}$ is $\bar{\mathbf{x}}$. These estimators are schematically represented on Fig. 1 for two examples of univariate pdfs.

2.2. Optimal nonlinear filtering assuming TG pdfs

In order to derive the TG filter, we first write the general filter equations, without any assumption on the shape of the error pdfs. The problem can be summarized as follows (Cohn, 1997).

Assume that we want to estimate a random vector, that is a function of time $t: \mathbf{x}(t), t \in [0, T]$. The input information of the estimation is: (i) an initial condition $\mathbf{x}^0$, with pdf $p[\mathbf{x}(0)|\mathbf{x}^0]$; (ii) a set of observation vectors $\mathbf{y}_k, k = 1, \dots, N$ at times $t_k \in [0, T], k = 1, \dots, N$, with pdfs $p[\mathbf{y}_k|\mathbf{x}(t_k)]$; (iii) probability distributions describing the future evolution of the system conditioned on the current state: $p[\mathbf{x}(t)|\mathbf{x}(t')], \forall t > t'$. From that information, it is possible to compute the posterior probability of the final state $p[\mathbf{x}(T)|\mathbf{x}^0, \mathbf{y}_k, k = 1, \dots, N]$. This defines the filtering problem, which can be used to initialize a subsequent forecast of the system using $p[\mathbf{x}(t)|\mathbf{x}(T)], t > T$.

The posterior probability distribution for the final state $p[\mathbf{x}(T)|\mathbf{x}^0, \mathbf{y}_k, k = 1, \dots, N]$ can be obtained by computing successively (assuming that the process is Markovian):

$$p[\mathbf{x}(t_1)|\mathbf{x}^0] = \int p[\mathbf{x}(t_1)|\mathbf{x}(0)] p[\mathbf{x}(0)|\mathbf{x}^0] d\mathbf{x}(0)$$

(First forecast of the pdf from $t = 0$ to $t = t_1$)

$$p[\mathbf{x}(t_1)|\mathbf{x}^0, \mathbf{y}_1] = p[\mathbf{x}(t_1)|\mathbf{x}^0] p[\mathbf{y}_1|\mathbf{x}(t_1)]$$

(First observational update at $t = t_1$)

:

$$p[\mathbf{x}(t_k)|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k-1]$$

$$= \int p[\mathbf{x}(t_k)|\mathbf{x}(t_{k-1})] p[\mathbf{x}(t_{k-1})|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k-1] d\mathbf{x}(t_{k-1})$$

(Forecast of the pdf from $t = t_{k-1}$ to $t = t_k$)

$$p[\mathbf{x}(t_k)|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k]$$

$$= p[\mathbf{x}(t_k)|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k-1] p[\mathbf{y}_k|\mathbf{x}(t_k)]$$

(Observational update at $t = t_k$)

:

$$p[\mathbf{x}(T)|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, N]$$

$$= \int p[\mathbf{x}(T)|\mathbf{x}(t_N)] p[\mathbf{x}(t_N)|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, N] d\mathbf{x}(t_N)$$

(Forecast of the pdf from $t = t_N$ to $t = T$)

In the rest of this section, we will show that, under certain assumptions, it is possible to build an optimal filter in which every pdf of the sequence is a truncated Gaussian pdf. Then, the pdfs can be written:

$$p[\mathbf{x}(t_k)|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k-1] = \mathcal{TN}_{\mathbf{x}(t_k)}(\tilde{\mathbf{x}}_k^f, \tilde{\mathbf{P}}_k^f; I_k)$$

$$p[\mathbf{x}(t_k)|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k] = \mathcal{TN}_{\mathbf{x}(t_k)}(\tilde{\mathbf{x}}_k^a, \tilde{\mathbf{P}}_k^a; I_k)$$

where $\tilde{\mathbf{x}}_k^f, \tilde{\mathbf{P}}_k^f$ and $\tilde{\mathbf{x}}_k^a, \tilde{\mathbf{P}}_k^a$ are the location vector and scale matrix of the truncated Gaussian pdf before and after the observational update at time t_k . (We have also allowed time varying inequality constraints, noted I_k for $t = t_k$).

2.2.1. Observational update of the pdf

The first step is to show that, during the observational update, the truncated Gaussian nature of the pdf is preserved: if $\mathbf{x}$ is a random vector with truncated Gaussian pdf: $p(\mathbf{x}) = \mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}^f, \tilde{\mathbf{P}}^f; I)$, with $I: \mathbf{A}\mathbf{x} \leq \mathbf{b}$, if $\mathbf{y}$ is a random vector conditioned on $\mathbf{x}$ with the Gaussian pdf: $p(\mathbf{y}|\mathbf{x}) = \mathcal{N}_{\mathbf{y}}(\mathbf{H}\mathbf{x}, \mathbf{R})$ (where $\mathbf{H}$ is any linear operator) then

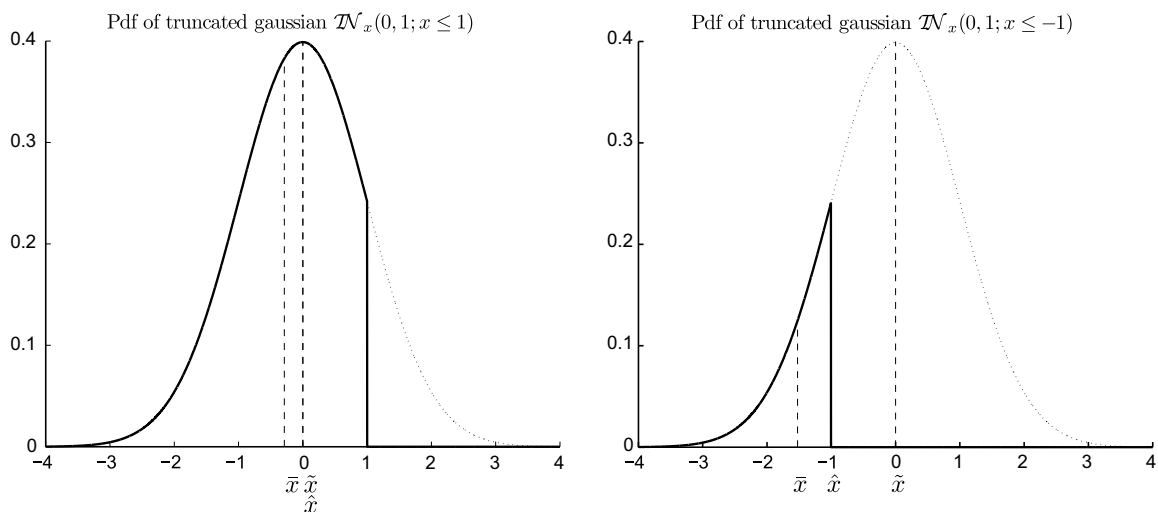


Fig. 1. Schematic representation of one-dimensional truncated Gaussian pdf (solid line), with their corresponding Gaussian pdf (dotted line). (The scaling with factor α is not represented in the figure.) Two typical cases are shown: on the left panel, the location vector $\tilde{\mathbf{x}} = 0$ of $\mathcal{TN}_{\mathbf{x}}(0, 1; \mathbf{x} \leq 1)$ fulfills the constraints so that the maximum likelihood estimator $\hat{\mathbf{x}} = \tilde{\mathbf{x}}$; on the right panel, the location vector $\tilde{\mathbf{x}} = 0$ of $\mathcal{TN}_{\mathbf{x}}(0, 1; \mathbf{x} \leq -1)$ does not fulfill the constraints so that $\hat{\mathbf{x}} = -1 \neq \tilde{\mathbf{x}}$. In both cases, the error variance minimizing estimator $\bar{\mathbf{x}}$ is different from $\tilde{\mathbf{x}}$ and $\hat{\mathbf{x}}$. Note however that this univariate example does not exhibit the full complexity of the general multivariate problem (several constraint planes delimiting a convex region of the state space).

the posterior probability of $\mathbf{x}$ conditioned on $\mathbf{y}$ is the truncated Gaussian pdf $p(\mathbf{x}|\mathbf{y}) = \mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}^a, \tilde{\mathbf{P}}^a; I)$ where

$$\tilde{\mathbf{x}}^a = \tilde{\mathbf{x}}^f + \mathbf{K}(\mathbf{y} - \mathbf{H}\tilde{\mathbf{x}}^f) \quad (7)$$

$$\tilde{\mathbf{P}}^a = (\mathbf{I} - \mathbf{K}\mathbf{H})\tilde{\mathbf{P}}^f \quad (8)$$

with

$$\tilde{\mathbf{K}} = \tilde{\mathbf{P}}^f \mathbf{H}^T (\mathbf{H} \tilde{\mathbf{P}}^f \mathbf{H}^T + \mathbf{R})^{-1} \quad (9)$$

Eqs. (7)–(9) are the classical formula for the observational update of the Kalman filter, with the difference that they are based on the location vector and scale matrix of the prior truncated Gaussian distribution and not on the mean and covariance of a prior Gaussian pdf. This property follows directly from the Bayes theorem (as demonstrated by Monin and Salut, 2001):

$$p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) = \mathcal{N}_{\mathbf{y}}(\mathbf{H}\mathbf{x}, \mathbf{R})\mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}^f, \tilde{\mathbf{P}}^f; I) \\ = \begin{cases} \frac{1}{2} \mathcal{N}_{\mathbf{y}}(\mathbf{H}\mathbf{x}, \mathbf{R})\mathcal{N}_{\mathbf{x}}(\tilde{\mathbf{x}}^f, \tilde{\mathbf{P}}^f) & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ 0 & \mathbf{A}\mathbf{x} > \mathbf{b} \end{cases}$$

This shows that the resulting distribution is either 0 or the product of the two normal pdf. Hence the solution is the Gaussian distribution $\mathcal{N}_{\mathbf{y}}(\mathbf{H}\mathbf{x}, \mathbf{R})\mathcal{N}_{\mathbf{x}}(\tilde{\mathbf{x}}^f, \tilde{\mathbf{P}}^f)$ truncated by the inequality constraints $I: \mathbf{A}\mathbf{x} \leq \mathbf{b}$; and it is already known (from the linear filter theory) that the parameters of this posterior Gaussian distribution are given by the classical formula (7)–(9). From the equation above (Bayes theorem), it is also easy to figure out that the same result is obtained if the Gaussian pdf that was assumed for $p(\mathbf{y}|\mathbf{x})$ is replaced by a truncated Gaussian pdf excluding some regions of the observation space, provided that these regions are already excluded through the inequality constraints I (working in the state space). Basically, the reason is that the same information about inequality constraints only needs to be taken into account once.

2.2.2. Forecast of the pdf

A general framework to forecast any pdf from time t_k to time t_{k+1} is to approximate the pdf forecast equation:

$$p[\mathbf{x}(t_{k+1})|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k] = \int p[\mathbf{x}(t_k)|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k] \\ p[\mathbf{x}(t_{k+1})|\mathbf{x}(t_k)] d\mathbf{x}(t_k) \quad (10)$$

using Monte Carlo integration. For instance, if we know that the random vector is governed by approximate dynamical laws:

$$\dot{\mathbf{x}} = f(\mathbf{x}, t) + \epsilon(t) \quad (11)$$

with known error pdf $p[\epsilon(t)]$, it is easy to obtain a sample of $p[\mathbf{x}(t_{k+1})|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k]$ by sampling an initial condition from $p[\mathbf{x}(t_k)|\mathbf{x}^0, \mathbf{y}_i, i = 1, \dots, k]$ and a model error from $p[\epsilon(t)]$, and integrating Eq. (11) for each initial condition and each model error in the sample. This corresponds exactly to the method used in the Ensemble Kalman Filter to propagate the pdf in time using an ensemble model forecast (Evensen, 2003).

In order to write the TG filter, it is assumed that, during the model forecast (Eq. (11)), the truncated Gaussian nature of the pdf is also preserved. This imposes quite a stringent condition to the nonlinear model dynamics $f(\mathbf{x}, t)$ and to the model noise $\epsilon(t)$, but this is the basic condition on which the TG filter relies. It means that if the model operator is applied to a sample of an initial TG pdf, the output sample can be considered as a sample of the output TG pdf. This is quite a strong assumption about the behaviour of a nonlinear model, but likely to be much better than the classical Gaussian assumption, because state vectors not satisfying the inequality constraints are now forced to zero probability. However, it is true that every feature of the forecasted pdf that does not fit the TG assumption will be lost. If the TG assumption is poor, important high order moment information may be destroyed and general nonlinear filtering is needed to de-

scribe correctly the probability distribution (see, for instance, Anderson, 2001; Budhiraja et al., 2007).

2.2.3. Summary of the algorithm

Using this assumption, the TG filter can thus be written as a sequence of four steps:

- (1) Sample $\mathcal{TN}_{\mathbf{x}(t_k)}(\tilde{\mathbf{x}}_k^a, \tilde{\mathbf{P}}_k^a; I_k)$: this first step requires an efficient algorithm to sample a truncated Gaussian distribution (see Section 2.3).
- (2) Compute a model forecast (with perturbations simulating the model error) for each element of the sample as initial condition: this corresponds to performing an ensemble forecast using the model equation (11).
- (3) Compute $\tilde{\mathbf{x}}_{k+1}^f$ and $\tilde{\mathbf{P}}_{k+1}^f$ from the forecast sample: this requires an algorithm to estimate the location vector and the scale matrix of a truncated Gaussian distribution from a sample (see Section 2.4).
- (4) Perform the observational update of $\mathcal{TN}_{\mathbf{x}(t_{k+1})}(\tilde{\mathbf{x}}_{k+1}^f, \tilde{\mathbf{P}}_{k+1}^f; I_k)$ to obtain $\tilde{\mathbf{x}}_{k+1}^a$ and $\tilde{\mathbf{P}}_{k+1}^a$, using Eqs. (7)–(9). For that purpose, no new tool need to be developed since Eqs. (7)–(9) are already implemented in Kalman filter applications.

Finally, from the sequence of pdfs described by their location vector and scale matrix, it is then possible to compute the maximum likelihood estimator ($\tilde{\mathbf{x}}_k^f$ and $\tilde{\mathbf{x}}_k^a$) using Eq. (6), or the error variance minimizing estimator ($\tilde{\mathbf{x}}_k^f$ and $\tilde{\mathbf{x}}_k^a$) using Eq. (3), together with the associated error covariance using Eq. (4). It must be stressed again that $\tilde{\mathbf{x}}^f$ and $\tilde{\mathbf{x}}^a$ do not necessarily satisfy the inequality constraints, and cannot be considered as useful estimators of the state of the system.

2.3. Sampling truncated Gaussian distributions

In the TG filter, sampling a truncated Gaussian distribution is required (i) in the forecast step, to propagate the pdf in time using the Monte Carlo method, and (ii) in order to compute the error variance minimizing estimator by computing integral (3) using Monte Carlo integrations.

Sampling the distribution $\mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$ is theoretically simple, since it is equivalent to sampling the Gaussian distribution $\mathcal{N}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}})$ eliminating the draws not satisfying the inequality constraints. However, this procedure (called ‘normal rejection sampling’) can be extremely inefficient if most of the draws from the Gaussian distribution must be eliminated.

A better method is the Gibbs sampler (Gelfand et al., 1992; Robert, 1995), that is a Monte Carlo Markov chain technique, using a sequence of draws from conditional distributions to characterize the joint distribution to be sampled. More precisely, in order to characterize the joint distribution $p(x_1, x_2, \dots, x_n)$, we draw successively from the conditional distributions $p(x_1|x_2, \dots, x_n)$, $p(x_2|x_1, \dots, x_n)$, ..., where the outputs of the $n-1$ previous draws are used to condition the next draws. It can be shown that the distribution of this Markov chain converges towards the joint distribution.

In our case, if $p(x_1, x_2, \dots, x_n) = \mathcal{TN}_{\mathbf{x}}(\mathbf{0}, \mathbf{I}; I)$, with $I: \mathbf{A}\mathbf{x} \leq \mathbf{b}$, then, the one-dimensional conditional pdfs $p(x_k|x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n)$ are the truncated Gaussian $\mathcal{TN}_{x_k}(\mathbf{0}, 1; I_k)$, with $I_k: A_{ik}x_k \leq b_i - \sum_{j \neq k} A_{ij}x_j$, $\forall i = 1, \dots, m$, where m is the number of inequality constraints (the size of the $\mathbf{b}$ vector). Indeed, the joint probability distribution is

$$p(\mathbf{x}) = p(x_1, x_2, \dots, x_n) = \mathcal{TN}_{\mathbf{x}}(\mathbf{0}, \mathbf{I}; I) \\ \propto \begin{cases} \exp\left(-\frac{1}{2} \sum_{i=1}^n x_i^2\right) & \text{for } \sum_{j=1}^n A_{ij}x_j \leq b_i, \quad \forall i = 1, \dots, m \\ 0 & \text{for } \sum_{j=1}^n A_{ij}x_j > b_i, \quad \forall i = 1, \dots, m \end{cases} \quad (12)$$

so that the one-dimensional conditional pdfs are

$$p(x_k | x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n) \propto \begin{cases} \exp(-\frac{1}{2}x_k^2) & \text{for } \sum_{j=1}^n A_{ij}x_j \leq b_i, \quad \forall i = 1, \dots, m \\ 0 & \text{for } \sum_{j=1}^n A_{ij}x_j > b_i, \quad \forall i = 1, \dots, m \end{cases} \quad (13)$$

The set of m inequality constraints $I_k : A_{ik}x_k \leq b_i - \sum_{j \neq k} A_{ij}x_j, \forall i = 1, \dots, m$ can be reduced to $x_k^{\min} \leq x_k \leq x_k^{\max}$ with

$$x_k^{\min} = \max_{A_{ik} < 0} \frac{b_i - \sum_{j \neq k} A_{ij}x_j}{|A_{ik}|} \quad \text{and} \quad x_k^{\max} = \min_{A_{ik} > 0} \frac{b_i - \sum_{j \neq k} A_{ij}x_j}{|A_{ik}|} \quad (14)$$

Obtaining $x_k^{\min} > x_k^{\max}$ means that the prescribed values of x_i ($i \neq k$) were impossible. The same problem occurs if $A_{ik} = 0$ with $b_i - \sum_{j \neq k} A_{ij}x_j < 0$.

Thus the Gibbs sampler algorithm reduces the problem to sampling one-dimensional truncated Gaussian distributions, a task for which an efficient algorithm has been proposed in Geweke (1991). The algorithm is based on the rejection method (described for instance in Press et al., 1992), using a uniform, normal, half-normal, or exponential function as a comparison distribution (as a function of $x_k^{\min}$ and $x_k^{\max}$, see Eq. (14)), in order to minimize the number of draws to be rejected. For instance, the exponential distribution is used as a comparison function in order to sample the tails of a truncated Gaussian distribution.

2.4. Computation of the TG parameters from a sample

Computing estimations of the parameters of Gaussian distributions is easy because the best estimation of the expected value and covariance of the distribution is directly related to the mean and covariance of the sample. The problem of estimating the location vector $\tilde{\mathbf{x}}$ and the scale matrix $\tilde{\mathbf{C}}$ of a truncated Gaussian distribution $\mathcal{TN}_{\mathbf{x}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$ from a sample $\mathbf{x}_1, \dots, \mathbf{x}_N$ (where N is the size of the sample) is much more difficult. In practice, a simple maximum likelihood estimator (see, for instance, Von Mises, 1964) can be used to compute $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{C}}$ maximizing the likelihood L :

$$L = \prod_{i=1}^N \mathcal{TN}_{\mathbf{x}_i}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$$

i.e. minimizing:

$$-\log L = \frac{1}{2} \sum_{i=1}^N (\mathbf{x}_i - \tilde{\mathbf{x}})^T \tilde{\mathbf{C}}^{-1} (\mathbf{x}_i - \tilde{\mathbf{x}}) + N \log \alpha$$

where α (function of $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{C}}$) is given by Eq. (2). A more rigorous (but more expensive) method to estimate the parameters of truncated Gaussian distributions is proposed by Griffiths (2006), who developed an approach to infer directly the posterior distribution of the location vector $\tilde{\mathbf{x}}$ and scale matrix $\tilde{\mathbf{C}}$: $p(\tilde{\mathbf{x}}, \tilde{\mathbf{C}} | \mathbf{x}_1, \dots, \mathbf{x}_N)$: a Gibbs sampler is used to compute a sample of this posterior distribution, from which best estimations of $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{C}}$ can be derived.

2.5. Order reduction

For many systems, it is known (or can be assumed) that the error probability density is only different from zero in a small subspace of the original state space. In such a situation, it is not worthy to simulate the error pdf in the full state space, since we only need it in the reduced dimension space in which it is not zero. In this section we show that the TG filter presented in the previous section can be easily adapted to work in a reduced dimension space. However, the definition of truncated Gaussian distribution given in Section 2.1 is valid only if $\tilde{\mathbf{C}}$ is not singular. In order to develop

a reduced rank filter, we will also need to work with ‘generalized’ truncated Gaussian distribution $\mathcal{TN}_{\mathbf{x}}^{\tilde{\mathbf{C}}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$, with singular $\tilde{\mathbf{C}}$ matrix, meaning that the probability is non zero only in the subspace defined by $\tilde{\mathbf{C}}$.

2.5.1. Observational update

If the error probability density is only significant in a small subspace of the original state space, it can be represented by $\mathcal{TN}_{\mathbf{x}(t_k)}^{\tilde{\mathbf{C}}}(\tilde{\mathbf{x}}_k^f, \tilde{\mathbf{P}}_k^f; I_k)$ (before the observational update at time t_k) and $\mathcal{TN}_{\mathbf{x}(t_k)}^{\tilde{\mathbf{C}}}(\tilde{\mathbf{x}}_k^a, \tilde{\mathbf{P}}_k^a; I_k)$ (after the observational update at time t_k), where $\tilde{\mathbf{P}}_k^f$ and $\tilde{\mathbf{P}}_k^a$ are singular:

$$\tilde{\mathbf{P}}_k^f = \tilde{\mathbf{S}}_k^f \tilde{\mathbf{S}}_k^{fT} \quad \text{and} \quad \tilde{\mathbf{P}}_k^a = \tilde{\mathbf{S}}_k^a \tilde{\mathbf{S}}_k^{aT}$$

Their rank (noted r_k) is not assumed the same for all k . It corresponds to the number of columns in the square root matrices $\tilde{\mathbf{S}}_k^f$ and $\tilde{\mathbf{S}}_k^a$. Such a square root representation of the error covariance matrices is naturally provided by ensemble methods. It can also be obtained from a principal component analysis of a sample of model states or from a time series that is assumed to characterize correctly the error covariance. Square roots can also be obtained from a full rank covariance matrix by a Cholesky decomposition or through the SVD algorithm.

The observational update can still be performed using formula (7) and (8). However, if $r_k \ll n$ and if the $\mathbf{R}$ matrix can be easily inverted, the computational complexity of the algorithm can be strongly reduced by computing successively (as in the SEEK filter, see Brasseur and Verron, 2006, for more details):

$$\mathbf{U}\mathbf{A}\mathbf{U}^T = (\mathbf{H}\tilde{\mathbf{S}}_k^f)\mathbf{R}^{-1}(\mathbf{H}\tilde{\mathbf{S}}_k^f)^T \quad (15)$$

$$\mathbf{K} = \tilde{\mathbf{S}}_k^f \mathbf{U}[\mathbf{I} + \Lambda]^{-1}(\mathbf{H}\tilde{\mathbf{S}}_k^f)^T \mathbf{R}^{-1} \quad (16)$$

$$\tilde{\mathbf{x}}^a = \tilde{\mathbf{x}}^f + \mathbf{K}(\mathbf{y} - \mathbf{H}\tilde{\mathbf{x}}^f) \quad (17)$$

$$\tilde{\mathbf{S}}^a = \tilde{\mathbf{S}}_k^f \mathbf{U}[\mathbf{I} + \Lambda]^{-1/2} \mathbf{U}^T \quad (18)$$

2.5.2. Forecast of the pdf

The forecast of the pdf can be performed by a sequence of four steps:

- (1) Sample $\mathcal{TN}_{\mathbf{x}(t_k)}^{\tilde{\mathbf{C}}}(\tilde{\mathbf{x}}_k^a, \tilde{\mathbf{P}}_k^a; I_k)$, where $I_k : \mathbf{A}_k \mathbf{x}(t_k) \leq \mathbf{b}_k$; this can be done by sampling $\mathbf{w}$ from the truncated Gaussian distribution $\mathcal{TN}_{\mathbf{w}}(\mathbf{0}; \mathbf{I}; I_k^w)$, where $I_k^w : \mathbf{A}_k \tilde{\mathbf{S}}_k^a \mathbf{w} \leq \mathbf{b}_k - \mathbf{A}_k \tilde{\mathbf{x}}_k^a$, and then computing $\mathbf{x}(t_k) = \tilde{\mathbf{x}}_k^a + \tilde{\mathbf{S}}_k^a \mathbf{w}$.
- (2) Compute a model forecast for each element of the sample as initial condition.
- (3) Diagnose the dimension r_{k+1} of the subspace necessary to contain all elements of the forecast sample (if it turns out that $r_{k+1} > r_k$, this may require to increase the size of the sample of $\mathcal{TN}_{\mathbf{x}(t_k)}^{\tilde{\mathbf{C}}}(\tilde{\mathbf{x}}_k^a, \tilde{\mathbf{P}}_k^a; I_k)$ until the forecasted subspace of dimension r_{k+1} is sufficiently sampled).
- (4) Compute $\tilde{\mathbf{x}}_{k+1}^f$ and $\tilde{\mathbf{S}}_{k+1}^f$ from the forecast sample. This amounts to computing the parameters of a truncated Gaussian for the components of $\mathbf{x}(t_{k+1})$ in a subspace of dimension r_{k+1} (using the algorithm described in Section 2.4).

In this algorithm, the dimension of the reduced dimension space in which the error pdf is represented is adaptive; this means that the algorithm can find by itself the directions of the state space along which no error is generated. It is also worthy to note that the computation of the maximum likelihood estimator and variance minimizing estimator (from the location vector and scale matrix) can be performed in the subspace of dimension r_k , using the truncated Gaussian pdf for the $\mathbf{w}$ vector defined as follows. If $\mathbf{x}$ is a random vector with pdf $\mathcal{TN}_{\mathbf{x}}^{\tilde{\mathbf{C}}}(\tilde{\mathbf{x}}, \tilde{\mathbf{C}}; I)$, with $I : \mathbf{A}\mathbf{x} \leq \mathbf{b}$ (for a singular scale matrix $\tilde{\mathbf{C}}$), and if $\tilde{\mathbf{C}} = \tilde{\mathbf{S}}\tilde{\mathbf{S}}^T$ (with all columns of $\tilde{\mathbf{S}}$ linearly independent), then the pdf of a random vector $\mathbf{w}$ such that

$\mathbf{x} = \tilde{\mathbf{x}} + \tilde{\mathbf{S}}\mathbf{w}$ is the truncated Gaussian pdf $\mathcal{TN}_{\mathbf{w}}(0, \mathbf{I}; I_w)$, where $I_w : \mathbf{A}\mathbf{S}\mathbf{w} \leq \mathbf{b} - \mathbf{A}\tilde{\mathbf{x}}$. This can reduce significantly the numerical cost of this operation.

2.6. Suboptimal filters

In many situations, good approximations of the optimal filter can be obtained from additional assumptions about the shape and behaviour of the error probability densities. Such approximations generally lead to reduce the computational complexity of the filter algorithm, and may be needed to make the problem numerically tractable. The computational complexity of the different versions of the algorithm is given in the [Appendix](#).

2.6.1. Quasi-normal approximation

If the inequality constraints only truncate far tails of the Gaussian distributions, they are seldom active, and do not bring much information into the system. Under this hypothesis, we can avoid the burden of estimating the truncated Gaussian parameters from the forecast ensemble, by setting:

$$\tilde{\mathbf{x}}^f \sim \tilde{\mathbf{x}}^f \quad \tilde{\mathbf{P}}^f \sim \mathbf{P}^f \quad (19)$$

Moreover, if the forecast of the scale matrix of this nearly Gaussian pdf is performed using the linear tangent model (assuming that the model is only weakly nonlinear), this amounts to approximating the optimal truncated Gaussian filter by an extended Kalman filter. The forecasted pdf is then fully determined by the computation of the location vector $\tilde{\mathbf{x}}^f \sim \tilde{\mathbf{x}}^f$ that can be approximated by applying the model equations to $\tilde{\mathbf{x}}^a$ or to $\tilde{\mathbf{x}}^a$.

In this quasi-normal approximation, the inequality constraints must still be explicitly taken into account by computing the $\tilde{\mathbf{x}}^a$ or $\tilde{\mathbf{x}}^a$ estimates (the mean or the maximum of the posterior truncated Gaussian pdf) as described in Section 2.1. This is a necessary step to ensure that the estimation verifies the constraints. In that way, the approximation can also be useful even if hypothesis (19) is not perfectly accurate. Note also that this operation is the only remaining algorithmic difference with respect to Gaussian filtering. If we choose to compute $\tilde{\mathbf{x}}^a$, this suboptimal filter is exactly the solution proposed in [Simon and Simon \(2005\)](#) to take into account inequality constraints in a Kalman filter.

2.6.1.1. Specified amplification of the scale matrix. Even with this approximation, most of the computational complexity of the algorithm remains in the forecasting of the truncated Gaussian pdf. However, the burden of forecasting the pdf can be avoided if it is observed (or can be assumed) that the effect of the model on the pdf is always to amplify the scale matrix as $\tilde{\mathbf{P}}_{k+1}^f = \frac{1}{\rho} \tilde{\mathbf{P}}_k^a$, with a known value of the amplification factor ρ . This corresponds to the approximation introduced by [Pham et al. \(1998\)](#) in their reduced-order Kalman filter (SEEK filter).

2.6.1.2. Asymptotic approximation of the scale matrix. The burden of forecasting the pdf can also be avoided if the matrix $\tilde{\mathbf{P}}_k^f$ is observed (or can be assumed) to exhibit a stationary or periodic asymptotic behaviour ($\tilde{\mathbf{P}}_k^f = \tilde{\mathbf{P}}_\infty^f$, where $\tilde{\mathbf{P}}_\infty^f$ is a constant or known periodic function of time).

2.6.2. Truncation of the error spectrum

In Section 2.5, we have presented an optimal reduced-order filter in which only the directions with non-zero error were dropped from the control space. In order to reduce the numerical cost of the algorithm, we can approximate that filter by keeping only a reduced set of directions containing a specified fraction of the total error variance. This can be performed by computing an EOF decomposition of the forecasted sample, and by defining the filter sub-

space as the space spanned by the first r_k EOFs explaining a fraction f of the total variance of the sample. This can save a lot of computation time because the sample needs only to be rich enough to sample correctly a subspace of dimension r_k (in order for the estimation of $\tilde{\mathbf{x}}_k^f$ and $\tilde{\mathbf{P}}_k^f$ from the sample to converge).

2.6.2.1. Reduced space of fixed dimension. A variant of the previous suboptimal filter would be to specify $r_k \forall k$ and diagnose the percentage of error variance that we loose. In this way, we have less control on the level of approximation in the filter, but more control on the numerical cost of the algorithm (that depends on r_k).

3. Application to a z-coordinate mixed layer model

3.1. Description of the ocean mixed layer model

In order to illustrate the method presented in Section 2, we will apply it to a one dimension (vertical) mixed layer model developed and validated in [Lacroix and Gregoire \(2002\)](#) and [Magri et al. \(2005\)](#). The model is based on primitive equations, that are reduced to their vertical dimension, assuming horizontal homogeneity. The model state variables are temperature (T), salinity (S), zonal velocity (U), meridional velocity (V) and turbulent kinetic energy. Potential density (ρ) is diagnosed from temperature and salinity using the equation of state, which is close to linear in the range of temperature and salinity spanned by the system. Vertical diffusivities and viscosities are computed from turbulent kinetic energy using a classical turbulent closure scheme. In practice, the equations are discretized using finite volumes of 5 m (between 0 and 400 m) and a time step of 1 h.

The model uses atmospheric data to calculate the surface boundary conditions like wind stress and heat flux, which determine the turbulent activity in the mixed layer. The surface tensions due to wind and heat fluxes are computed with bulk formula, using real meteorological conditions (as in [Magri et al., 2005](#)). Meteorological data come from a station located near Nice, France (7.13°E, 43.38°N). Solar radiation was measured every hour, wind (intensity and direction) every 3 h, air temperature, clouds, relative humidity and atmospheric pressure every 6 h, and precipitation every 24 h.

A reference interannual simulation was performed for the years 1985–1988. [Fig. 2](#) shows the behaviour of the model on the temperature and potential density variables over the 4 years of the simulation. With regard to our application, a key feature of the simulations with such a model is that the water column always remains (almost) hydrostatically stable. Neglecting the thermobaric effects, this means that the potential density (here referenced to the surface) always increases monotonically with depth (see [Fig. 2](#)). This introduces the inequality constraints I in the ensemble of possible state vectors:

$$I : \rho_k - \rho_{k+1} \leq \epsilon, \quad \forall k \text{ (increasing with depth)} \quad (20)$$

so that state error probability distribution cannot be strictly Gaussian in this case study. In the inequality constraints I (Eq. 20), a small tolerance for density inversions $\epsilon = 0.01 \text{ kg/m}^3$ has been introduced, because the model sometimes produces slightly unstable profiles (thus violating its own prior hypotheses). Such tolerance is thus necessary, because the TG filter cannot work properly if the inequality constraints are not satisfied by the model forecasts themselves. This would violate our basic assumption.

3.2. Description of the twin experiments

The numerical experiments are designed to focus the study on the observational update which represents the main difference

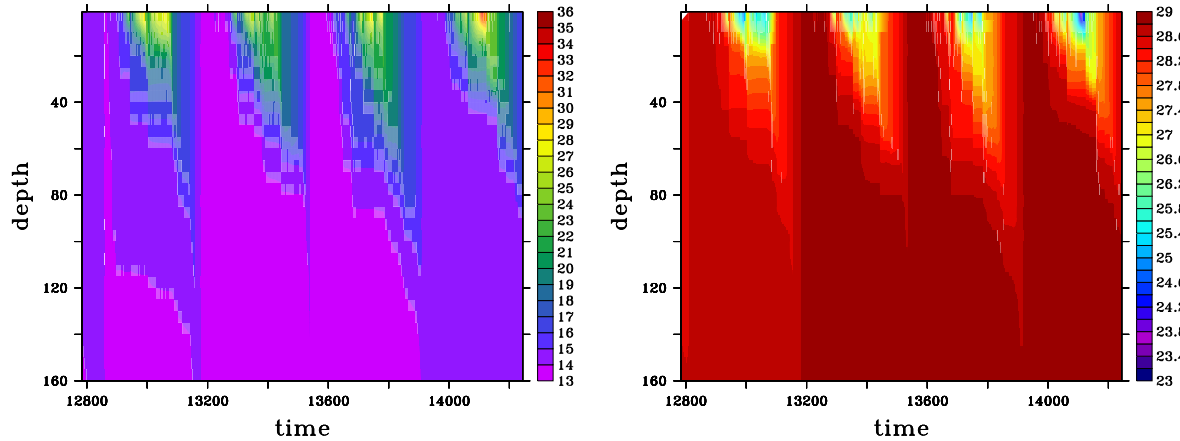


Fig. 2. Reference simulation from January 1, 1985 to December 31, 1988: temperature ($^{\circ}\text{C}$, on the left); potential density (kg/m^3 , on the right). Time is in Julian days (using January 1, 1950 as reference date).

with respect to other filters like the ensemble Kalman filter. The experiments are performed according to the following protocol in five steps: (i) make an assumption about the sources of error in the system and postulate a prior probability distribution for these errors (Section 3.2.1), (ii) sample this prior distribution to perform an ensemble model forecast (Section 3.2.1), (iii) define a reference simulation, from which the observations will be extracted, and a perturbed simulation, in which the observations will be assimilated (Section 3.2.2), (iv) parameterize the forecast error pdf from the ensemble forecast, either as a Gaussian or as a truncated Gaussian pdf: a reduced rank parameterization will be used in the subspace defined by the first EOFs of the ensemble (Section 3.3), and (v) compare the Gaussian and the truncated Gaussian observational updates (Section 4).

3.2.1. Ensemble generation

In order to set up twin experiments, we assume that the only source of error in the system comes from the atmospheric data (wind and air temperature only), with known probability distribution. We make the following assumptions.

- Errors on the amplitude and directions of the wind are independent, following respectively a Gaussian distribution (ϵ_W) and a uniform distribution (Θ).
- Errors on air temperature follow a Gaussian probability law (ϵ_{T_a}).

Errors $\delta u, \delta v, \delta T_a$ on wind components and air temperature can thus be expressed as follows:

$$\delta u = \Delta W * \epsilon_W * \cos(\Theta) \quad (21)$$

$$\delta v = \Delta W * \epsilon_W * \sin(\Theta) \quad (22)$$

$$\delta T_a = \Delta T_a * \epsilon_{T_a} \quad (23)$$

where $\Theta \sim \mathcal{U}(0, 2\pi)$; $\epsilon_W \sim \mathcal{N}(0, 1)$; $\epsilon_{T_a} \sim \mathcal{N}(0, 1)$, and $\Delta W, \Delta T_a$ are the amplitude of the perturbations on wind and air temperature. In our experiment, the amplitude of the perturbation is set to $\Delta W = 4 \text{ m s}^{-1}$ and $\Delta T_a = 1^{\circ}\text{C}$.

From this distribution, we can sample an ensemble of atmospheric data and generate the corresponding ensemble of model simulations (Monte Carlo simulations), as represented schematically in Fig. 3. The reference central simulation is the solution of the model using the real forcing data. The other members of the ensemble start from the same initial conditions (on October 8, 1985), but with perturbations (Eqs. (21)–(23)) added to the atmospheric data. This ensemble reflects the sensitivity of the model

forecast to errors in the forcing data. It is a sample of the pdf of the resulting error in the model forecast. One of them is the forecast of the twin experiments, i.e. the perturbed free model solution that we want to improve by data assimilation. Fig. 4 illustrates the 10-day ensemble forecast (1000 members) for density, temperature, salinity and turbulent kinetic energy. It can easily be verified that all members of the ensemble satisfy the inequality constraints I (hydrostatic stability of the water column), since they are all solutions of the model equations. Results below 100 m are not shown in the figure (as in all following figures) because wind and air temperature perturbations have a negligible effect on the deeper layers.

In order to choose the size of the sample to be used in our experiments, we analyze the convergence of the sample distribution as a function of sample size. For instance, Fig. 5 shows the effect of sample size on the distribution of surface temperature and surface density. We observe that, as sample size increases, the distribution converges towards that obtained for the largest sample (60,000 members). For the purpose of this study, it will be sufficient to use a sample of 1000 members to represent the model forecast pdf, considering that it already represents the model behaviour fairly well.

3.2.2. Set up of the experiment

In order to perform twin experiments, we need to define a reference simulation, considered as the truth, from which observations will be extracted, and a perturbed simulation, affected by the errors, in which observations will be assimilated. In our experiment, both the reference simulation and the perturbed simulation are sampled from the distribution described in Section 3.2.1 (see Fig. 3). However, only the perturbed simulation belongs to the 1000-member sample that will be used to parameterize the forecast error distribution in the assimilation experiment. It should also be noted that these two simulations were not chosen randomly, but selected in order to make the problem of inequality constraints most apparent, as discussed below.

Fig. 6 compares the evolution of density in the reference and perturbed simulations for a 10-day forecast starting on October 8, 1985. In the reference simulation, the mixed layer deepens during this 10-day period due to a strong wind blowing at the surface. In the perturbed simulation, the wind is somewhat weaker, so that the deepening of the mixed layer is slower: the water column remains more stratified than in the reference simulation.

In this situation, it can be expected that the effect of the observational update will be to unstratify the water column, in order to

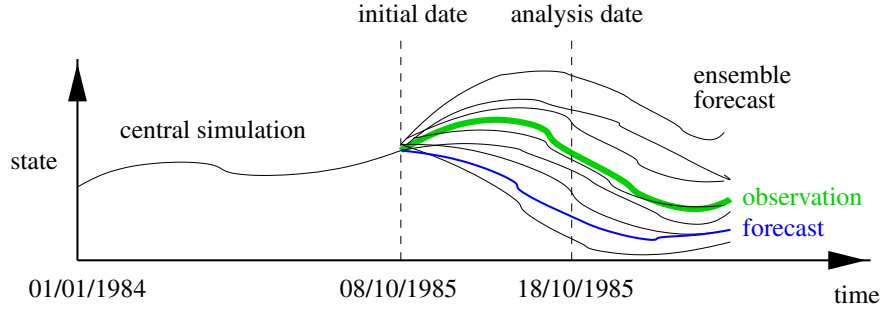


Fig. 3. Scheme of the ensemble experiment: from October 8, 1985, an ensemble forecast is produced by simulating errors in the atmospheric data. From that ensemble, we select a reference simulation (the truth, from which the observations are extracted: green line), a perturbed simulation (the forecast in which the observations will be assimilated: blue line), and a 1000-member sample (not including the truth) from which the forecast error pdf will be parameterized (black lines).

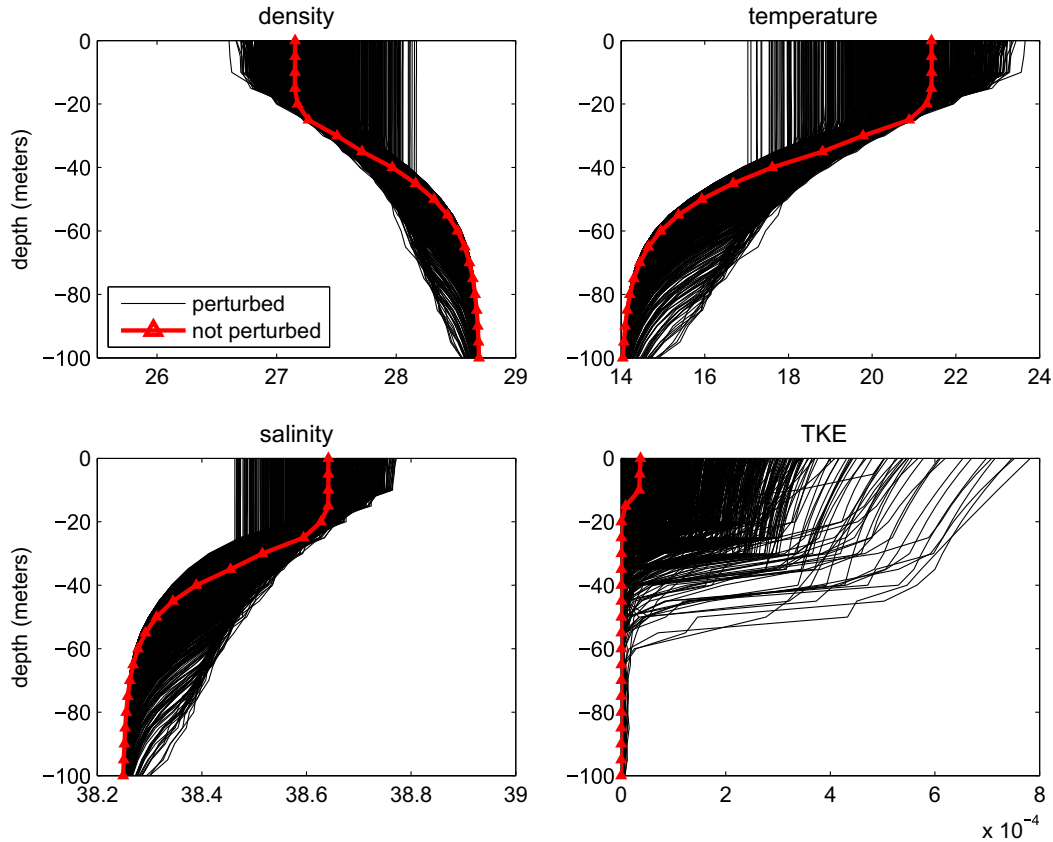


Fig. 4. 10-day ensemble forecast for potential density (kg/m^3), temperature ($^{\circ}\text{C}$), salinity (psu) and turbulent kinetic energy ($\text{m}^2 \text{s}^{-2}$). 1000 members are represented in the figure. The red line represents the central simulation (without perturbations of the atmospheric data).

get closer to the reference state. It is in this kind of situation that unstable water columns are most likely to show up in the estimated state, if nothing is done to impose inequality constraints. For this reason, the method described in Section 2 will be tested by analysing the performance of an observational update on October 18, 1985, as compared with the classical Gaussian analysis. This will be done for several types of observation operators and for different levels of observational error (see Section 4.2).

3.3. Parameterization of the forecast error pdf from the ensemble experiment

In order to estimate the state of the system on October 18, 1985 from the observations, we need a parameterization of the forecast

error pdf, either as a Gaussian pdf (to use the classical least-squares scheme) or as a truncated Gaussian pdf (to assess the new scheme presented in Section 2).

As a Gaussian parameterization of the forecast error pdf, we will use $\mathcal{N}(\mathbf{x}^f, \mathbf{P}^f)$, where $\mathbf{x}^f$ is the forecast state and $\mathbf{P}^f = \mathbf{S}^f \mathbf{S}^{fT}$ is a reduced-order representation of the covariance of the 1000-member ensemble described in Section 3.2.1. $\mathbf{S}^f$ is a $n \times r$ matrix, whose r columns are the r first EOFs of the ensemble. Fig. 7 (left panels) shows a sample of this distribution as computed with $r = 3$ and $r = 7$ (only potential density profiles are shown). From this figure, it is apparent that many profiles of the sample are hydrostatically unstable, thus physically unacceptable. This already suggests that the Gaussian assumption is not a very good way to represent the forecast error pdf.

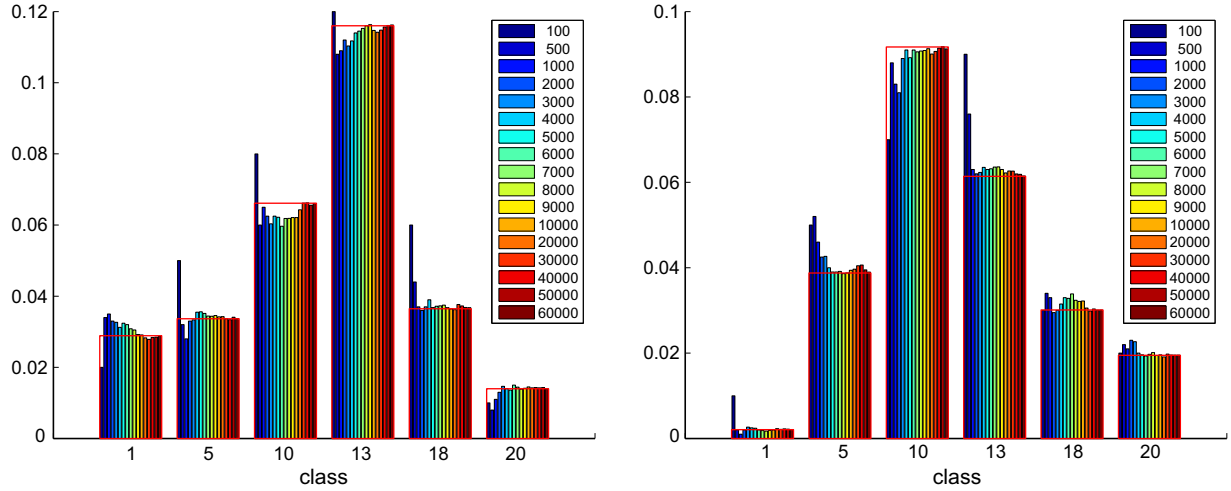


Fig. 5. Convergence of the sample distribution as a function of sample size, for surface temperature (left panel) and surface density (right panel). The distribution is approximated by a histogram of 20 classes (using interval widths of 0.28 °C, between 17.8 and 23.2 °C, for temperature and 0.07 kg/m³, between 26.7 and 28.1 kg/m³, for density). Only six of these 20 classes (number 1, 5, 10, 13, 18 and 20) are shown on the figure. Samples contain from 100 to 60 000 members. They must be compared to the most representative sample of 60 000 members (represented by the red bar furthest to the right).

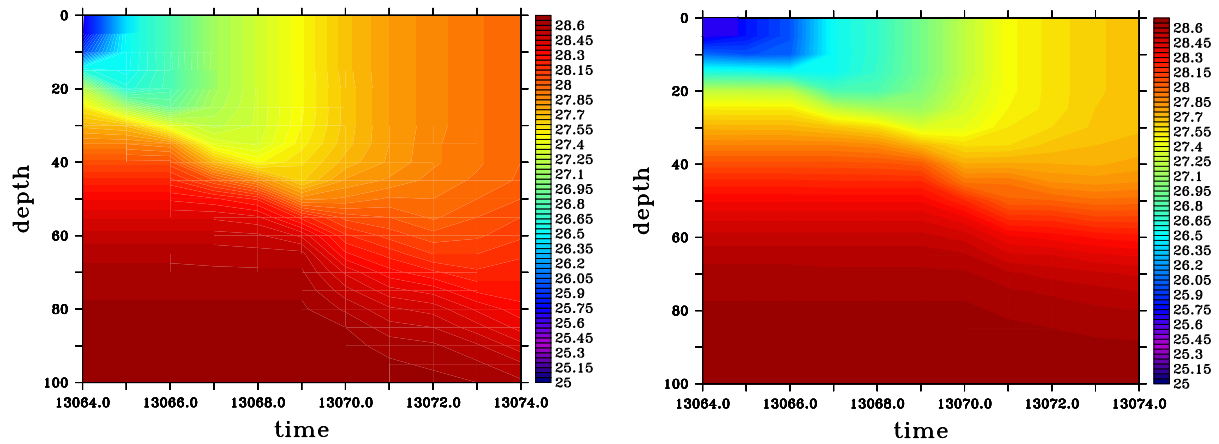


Fig. 6. Evolution of density between October 8 and October 18, 1985 in the reference simulation (true state) on the left, and in the perturbed simulation (false state) on the right. (Time is in Julian days.)

As a truncated Gaussian parameterization of the forecast error pdf, we use $\mathcal{TN}(\mathbf{x}^f, \mathbf{P}^f; I)$ with inequality constraints I given by Eq. (20). This means that, for the sake of simplicity, we make use of the approximation described in Section 2.6.1, and use the same elements $\mathbf{x}^f$ and $\mathbf{P}^f$ to parameterize the Gaussian and truncated Gaussian pdf (quasi-normal approximation). Fig. 7 (right panels) shows a sample of this distribution as computed with $r = 3$ and $r = 7$. The sample was obtained using a Gibbs sampler (as described in Section 2.3). Note that the tests we made (not shown) to estimate the truncated Gaussian parameters from a sample of the distribution (using the algorithm described in Section 2.4) suggest that this does not have an important impact on the results in our case study.

As expected, all members of the ensemble satisfy the inequality constraints, and are thus physically acceptable. This does not mean that $\mathcal{TN}(\mathbf{x}^f, \mathbf{P}^f; I)$ is necessarily a good approximation of the true forecast error pdf, but at least these basic constraints are satisfied by the pdf. Using the quasi-normal approximation means that the only difference with respect to the Gaussian assumption is that zero probability is forced in prohibited regions of the state space. Therefore, this approximation is necessarily better than the Gaussian assumption.

To illustrate why the Gaussian assumption is not very appropriate in this case study, Fig. 8 (left panel) shows the distribution of

differences in potential density between the seventh and first level of the model, as described by the 1000-member ensemble forecast. This variable is never negative and probability density is large close to zero; such a discontinuity in probability density cannot be approximated well by a Gaussian pdf $\mathcal{N}(\mathbf{x}^f, \mathbf{P}^f)$ (middle panel of Fig. 8). On the other hand, the right panel, representing a sample of the truncated Gaussian distribution $\mathcal{TN}(\mathbf{x}^f, \mathbf{P}^f; I)$, shows a distribution that is much more similar to the ensemble forecast. It should, however, be stressed that the forecast pdf is not a truncated Gaussian pdf either. For example, the peak located in the neighbourhood of the constraints (230 members close to zero on the left panel) is not well reproduced by the truncated Gaussian sample. The purpose of this section is only to show that the truncated hypothesis provides a more consistent approximation of the forecast pdf than the Gaussian hypothesis.

4. Observational update of a truncated Gaussian pdf: sensitivity studies

The objective of this section is to apply the observational update presented in Section 2 to the case study described in Section 3. In Section 4.1, a first example is presented for one particular case study using a specific observation system and a given size r of

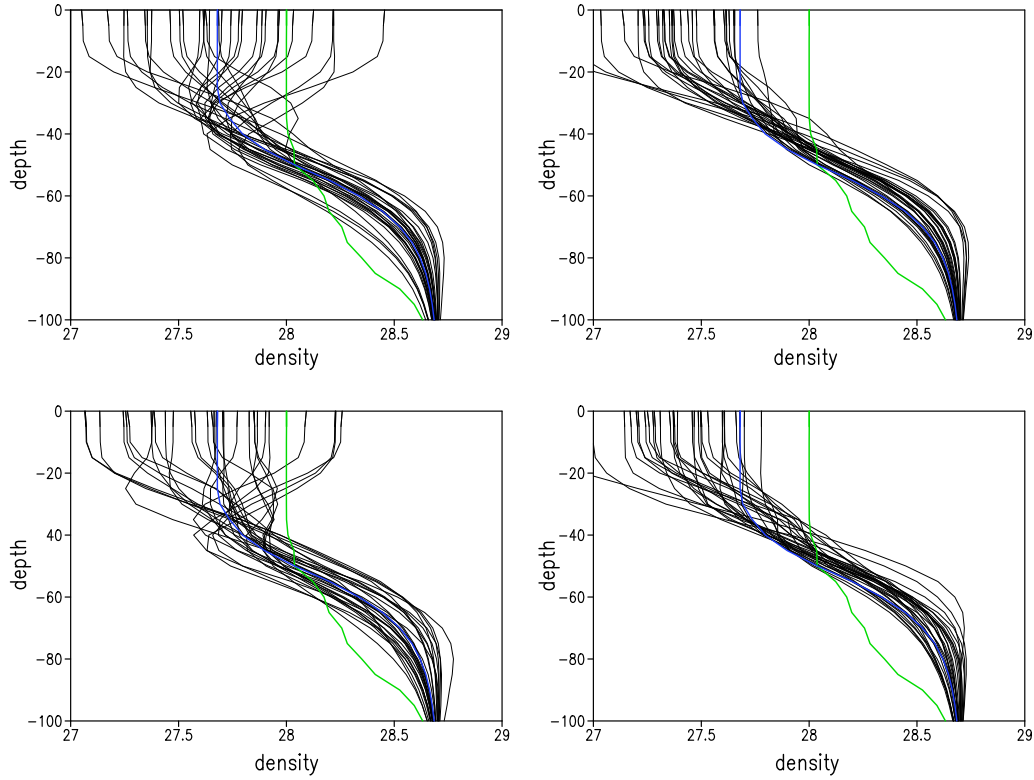


Fig. 7. Samples (30 members) of the Gaussian distribution $\mathcal{N}(\mathbf{x}^f, \mathbf{P}^f)$ for $r = 3$ (top left panel) and $r = 7$ (bottom left panel) and of the truncated Gaussian distribution $\mathcal{TN}(\mathbf{x}^f, \mathbf{P}^f; I)$ for $r = 3$ (top right panel) and $r = 7$ (bottom right panel). The forecast state $\mathbf{x}^f$ is in blue and the reference simulation (the truth) is in green. Only potential density profiles are shown.

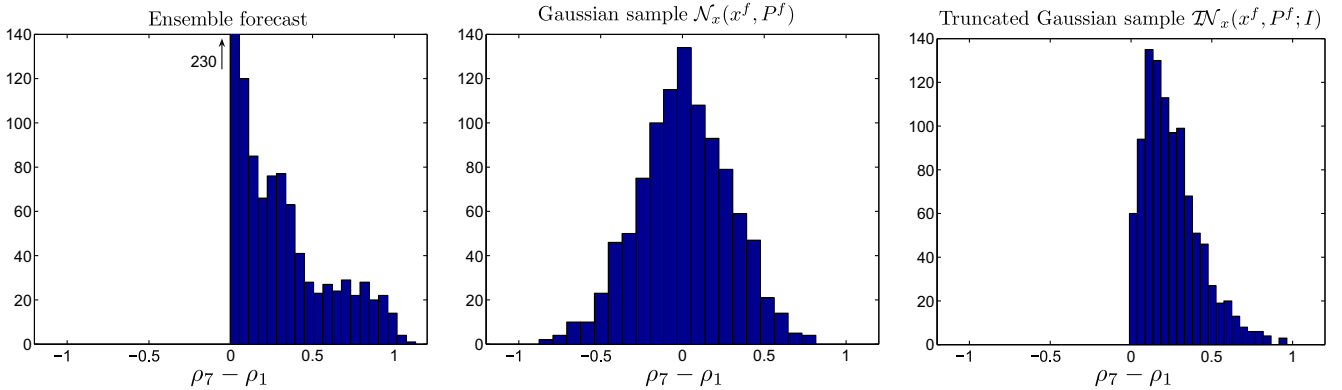


Fig. 8. Histograms of the difference between potential density (kg/m^3) at level 7 and level 1 comparing the ensemble forecast (left panel), a sample of the Gaussian distribution $\mathcal{N}(\mathbf{x}^f, \mathbf{P}^f)$ for $r = 7$ (middle panel), and in a sample of the truncated Gaussian distribution $\mathcal{TN}(\mathbf{x}^f, \mathbf{P}^f; I)$ for $r = 7$ (right panel). All samples contain 1000 members. In this example, the truncated Gaussian hypothesis represents a better approximation of the ensemble forecast behaviour than the Gaussian hypothesis.

the error subspace, with the aim of describing in detail how the algorithm actually works. Section 4.2 compares several experiments with a view to analysing the effect of different types of observation system on the truncated Gaussian observational update. More specifically, the aim is to study under what circumstances the truncated Gaussian hypothesis provides better results than those obtained with classical analysis. In Section 4.3, we analyze the effect of using a reduced space with an insufficient size r .

4.1. Example of observational update

In this section, we first present an ensemble experiment in order to show the consequences of working with a truncated Gaussian hypothesis as opposed to a Gaussian hypothesis. The case study is as follows:

- Observations of temperature and salinity assimilated every 10 grid points (i.e. every 50 m from 0 to 400 m depth).
- Observation error standard deviation for temperature and salinity: $\sigma_{\text{TEM}} = 0.3^\circ\text{C}$; $\sigma_{\text{SAL}} = 0.05$ psu.
- The forecast error pdf (illustrated in Fig. 7) is described in a reduced space of dimension $r = 7$.

Using these settings, we will perform the observational update of the pdf according to the method described in Section 2. We will successively assume that the forecast error is a Gaussian pdf (Fig. 7, bottom left panel) and a truncated Gaussian pdf (Fig. 7, bottom right panel).

The observational update of the Gaussian pdf $\mathcal{N}(\mathbf{x}^f, \mathbf{P}^f)$ is the Gaussian pdf $\mathcal{N}(\mathbf{x}^a, \mathbf{P}^a)$ where $\mathbf{x}^a$ and $\mathbf{P}^a$ are computed from $\mathbf{x}^f$ and $\mathbf{P}^f$ using Eqs. (7)–(9) (in our case, Eqs. (15)–(18) since $\mathbf{P}^f$ is a

reduced-order covariance matrix). Fig. 9 (left panel) shows a sample of $\mathcal{N}(\mathbf{x}^a, \mathbf{P}^a)$, that should be compared to the sample of $\mathcal{N}(\mathbf{x}^f, \mathbf{P}^f)$ in Fig. 7 (bottom left panel). In the Gaussian case, the error variance minimizing estimator and the maximum likelihood estimator are both equal to $\mathbf{x}^a$, shown in red in Fig. 9. This analysis typically gives an unstable density profile, which is physically unrealistic. Restarting the model from such an unstable initial condition will clearly generate a lot of spurious mixing to restore hydrostatic stability. This is a typical problem with this model if the inequality constraints are not taken into account. Note also that most profiles of the analysis covariance error sample are unstable, which means that the estimated error pdf is not physically realistic and could not be exploited to initialize an ensemble forecast. More intuitively, the effect of the Gaussian observational update is basically to transform the prior Gaussian pdf into a new Gaussian pdf with larger probability close to the observation. The problem is precisely that without explicit constraint, most sample elements of the updated Gaussian are unstable, because most elements of the prior pdf that are close to the observation are unstable. Again this is the result of the invalid prior Gaussian assumption.

The observational update of the truncated Gaussian pdf $\mathcal{TN}(\mathbf{x}^f, \mathbf{P}^f; I)$ is the truncated Gaussian pdf $\mathcal{TN}(\mathbf{x}^a, \mathbf{P}^a; I)$ where $\mathbf{x}^a$ and $\mathbf{P}^a$ are again computed from $\mathbf{x}^f$ and $\mathbf{P}^f$ using Eqs. (15)–(18). A sample of $\mathcal{TN}(\mathbf{x}^a, \mathbf{P}^a; I)$ can be computed using a Gibbs sampler, as explained in Section 2.3. Fig. 9 (right panel) shows such a sample, that should be compared to the sample of $\mathcal{TN}(\mathbf{x}^f, \mathbf{P}^f; I)$ in Fig. 7 (bottom right panel). As explained in Section 2, the error variance minimizing estimator $\bar{\mathbf{x}}^a$ (red profile in Fig. 9, right panel) can be obtained by computing (using Monte Carlo integration) the mean of such a sample ($\bar{\mathbf{x}}^a$ has been computed with a sample of much more than the 30 members represented in the figure, in order to reach a sufficient accuracy). As can be observed in Fig. 9, the method enables us to obtain the required vertical hydrostatic stability, thus providing us with a realistic initial condition to restart the model. In this case, the RMS difference between analysis and reference states along the vertical column is also better with the truncated Gaussian hypothesis.

In Fig. 9, the sample of $\mathcal{TN}(\mathbf{x}^a, \mathbf{P}^a; I)$, on the right panel, could have been obtained by sampling $\mathcal{N}(\mathbf{x}^a, \mathbf{P}^a)$ and rejecting all profiles not satisfying the constraints (as explained in Section 2.3). However, the sample of $\mathcal{N}(\mathbf{x}^a, \mathbf{P}^a)$ shown in Fig. 9 (left panel) suggests that, in this case, the proportion of stable profiles is so small, that this method would not be efficient enough to be applicable. A Gibbs sampler (described in Section 2.3) is needed to produce a sample of $\mathcal{TN}(\mathbf{x}^a, \mathbf{P}^a; I)$ at an affordable numerical cost.

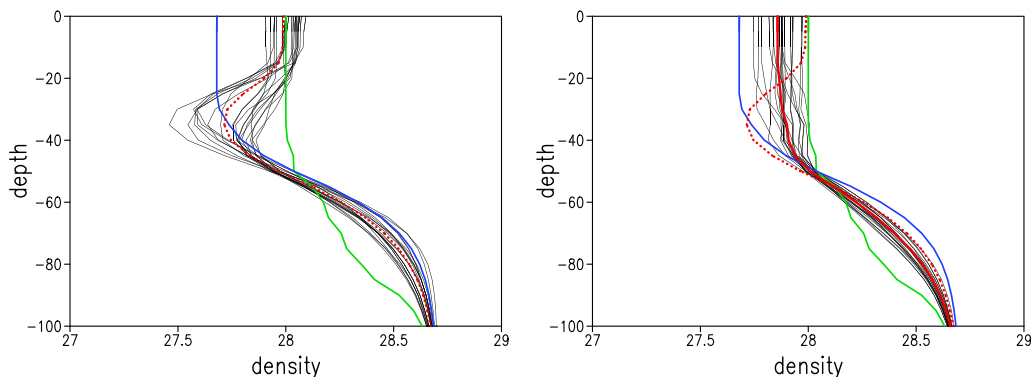


Fig. 9. Comparison of observational updates using the Gaussian assumption (left panel) and the truncated Gaussian assumption (right panel), as illustrated by samples (30 members) of the posterior distributions $\mathcal{N}(\mathbf{x}^a, \mathbf{P}^a)$ and $\mathcal{TN}(\mathbf{x}^a, \mathbf{P}^a; I)$ for $r = 7$. The figure compares the truncated Gaussian variance minimizing estimator $\bar{\mathbf{x}}^a$ (solid red line on the right panel) with the Gaussian estimator $\mathbf{x}^a$ (dotted red line), the forecast state $\mathbf{x}^f$ (blue line) and the truth (green line). Note that with the quasi-normal assumption $\bar{\mathbf{x}}^a$ is also the location vector of the truncated Gaussian pdf.

This case study was meant to illustrate the problems encountered when assuming a Gaussian hypothesis. Nevertheless, it could be argued that the behaviour of the solution may depend on the specific settings of the experiment. This question is examined in the following sections by looking at the sensitivity of the results as a function of three factors: the quantity of assimilated observations, the observation error and the dimension r of the reduced space.

4.2. Sensitivity to the observation system

To study the sensitivity of the results to the observation system, the experiments described in the previous section were reproduced for three different observation systems and three different values of the observational error. The three observation systems considered are:

1. Temperature and salinity observed at all levels of the water column (every 5 m from 0 to 400 m depth, i.e. 81 observations of temperature and 81 observations of salinity).
2. Temperature and salinity observed at 10-level intervals in the water column (every 50 m from 0 to 400 m depth, i.e. 9 observations of temperature and 9 observations of salinity).
3. Temperature and salinity observed at surface level only.

For each observation system, we used three types of observational error standard deviation (σ_{TEM} for temperature and σ_{SAL} for salinity):

- a. Observations are very accurate: $\sigma_{\text{TEM}} = 0.01$ °C; $\sigma_{\text{SAL}} = 0.01$ psu.
- b. Observations are moderately accurate: $\sigma_{\text{TEM}} = 0.3$ °C; $\sigma_{\text{SAL}} = 0.05$ psu.
- c. Observations are not accurate: $\sigma_{\text{TEM}} = 1$ °C; $\sigma_{\text{SAL}} = 0.2$ psu.

The results, computed with $r = 7$, are summarized in Figs. 10 and 11, which display the same information as Fig. 9 for all 9 observation systems.

Consider first the observation system with accurate observations at all levels of the water column (case 1.a). In this case, the Gaussian estimator $\mathbf{x}^a$ (dotted red line) and the truncated Gaussian estimator $\bar{\mathbf{x}}^a$ (red line) are both indistinguishable from the reference state $\mathbf{x}^f$ (green line). Moreover, all members of the sample of the posterior distributions $\mathcal{N}(\mathbf{x}^a, \mathbf{P}^a)$ and $\mathcal{TN}(\mathbf{x}^a, \mathbf{P}^a; I)$ are very close to $\bar{\mathbf{x}}^a$. So much information about the system is available that both estimators are very close to the truth, with a very small

associated error bar. In such a situation, taking into account the inequality constraints does not influence the estimation much, because all necessary information is already available through the observation system.

Second, if we maintain a good level of observation accuracy (as in case 1.a), but reduce the number of observations on the vertical (cases 2.a and 3.a), we observe that the Gaussian estimator produces an unstable density profile, with a large discrepancy with respect to the truth where no observations are available. On the other hand, the truncated Gaussian estimator remains very close to the truth in cases 2.a and 3.a. In this situation, the inequality constraints provide enough information to compensate for the lack of temperature and salinity observations. Accordingly, the spread of the posterior truncated Gaussian pdf (visualized by a 30-member sample) remains moderate, while the spread of the posterior Gaussian pdf $\mathcal{N}(\mathbf{x}^a, \mathbf{P}^a)$ becomes much larger.

Third, if we maintain an observation system with temperature and salinity observations at all levels (as in case 1.a) but reduce observation accuracy (cases 1.b and 1.c), we observe that the spread of the posterior pdf increases as observation accuracy decreases. Accordingly, both the Gaussian and truncated Gaussian estimators separate from the truth, and the risk of an unstable Gaussian estimator being produced (as in case 1.c) increases. However, in this case the inequality constraints do not provide enough

information to compensate for the lack of accuracy in temperature and salinity observation. In between the forecast state $\mathbf{x}^f$ (blue line) and the truth (green line), there exists a range of stable profiles (satisfying the inequality constraints) that are compatible with observation accuracy. Note that in cases 1.a, 2.a, 3.a, there were none, which is why the inequality constraints provided sufficient additional information to make the estimator so accurate.

Finally, if we consider the remaining cases (2.b, 2.c, 3.b, 3.c), it may be observed that as a general rule the less the amount of information provided by the observation system, the greater the spread of the posterior pdf, so that it becomes more important to take the inequality constraints into account in the estimation process. Hence, the method proposed in this paper is likely to be most useful in problems for which input information is incomplete or not totally accurate.

4.3. Sensitivity to the size of the reduced space

To analyze what happens if the size r of the reduced space is insufficient to describe the subspace in which probability density is significant, it is enough to reduce the size of the reduced space from $r = 7$ to $r = 3$ and to analyze the effect on the observational update using observation system 1.a described in Section 4.2 (temperature and salinity observed at all levels of the water column

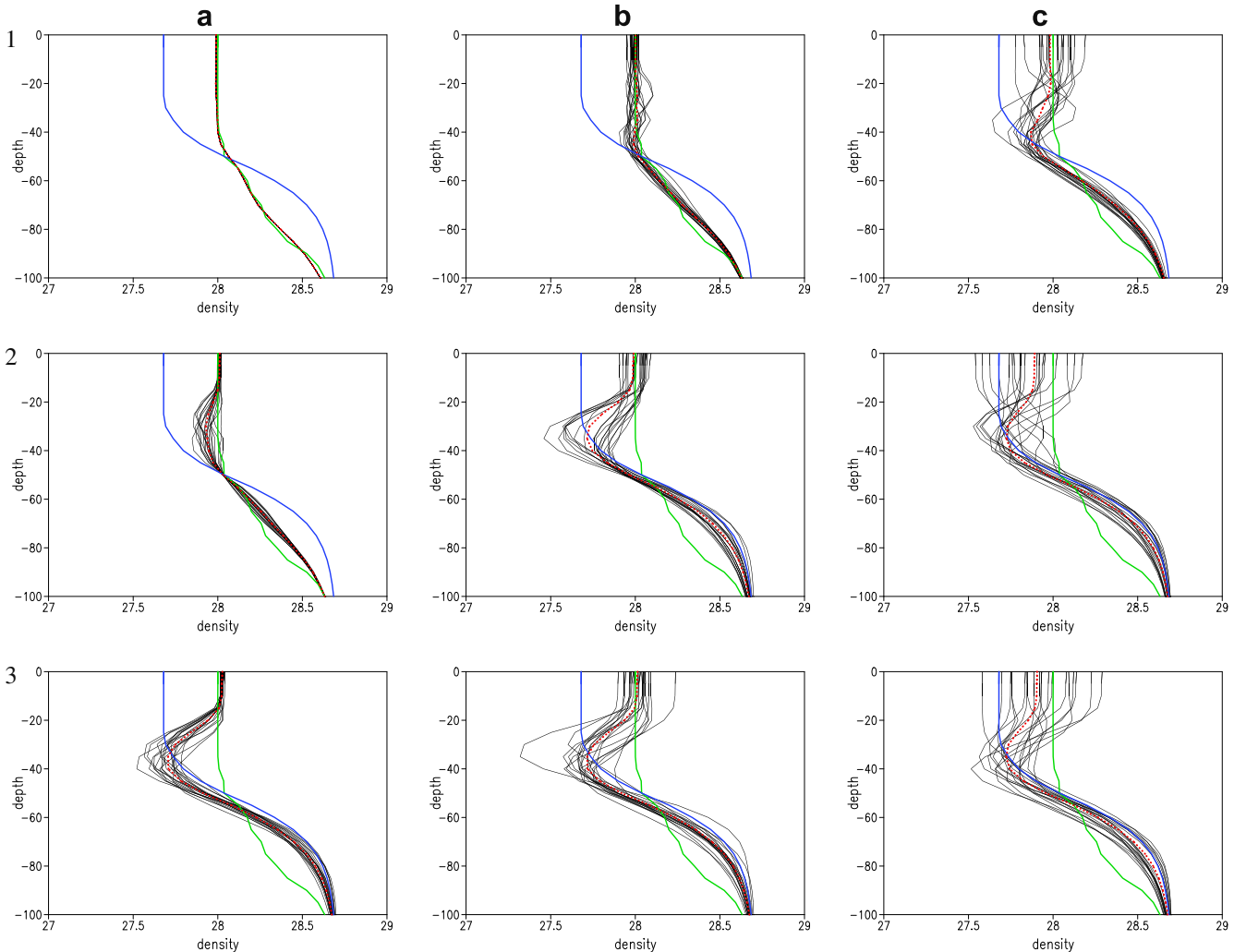


Fig. 10. Sample (30 members) of the Gaussian $\mathcal{N}(\mathbf{x}^a, \mathbf{P}^a)$ for 9 different observation systems and $r = 7$. The figure also shows the forecast state $\mathbf{x}^f$ (blue line), the true state $\mathbf{x}^t$ (green line) and the Gaussian estimator $\mathbf{x}^a$ (dotted red line).

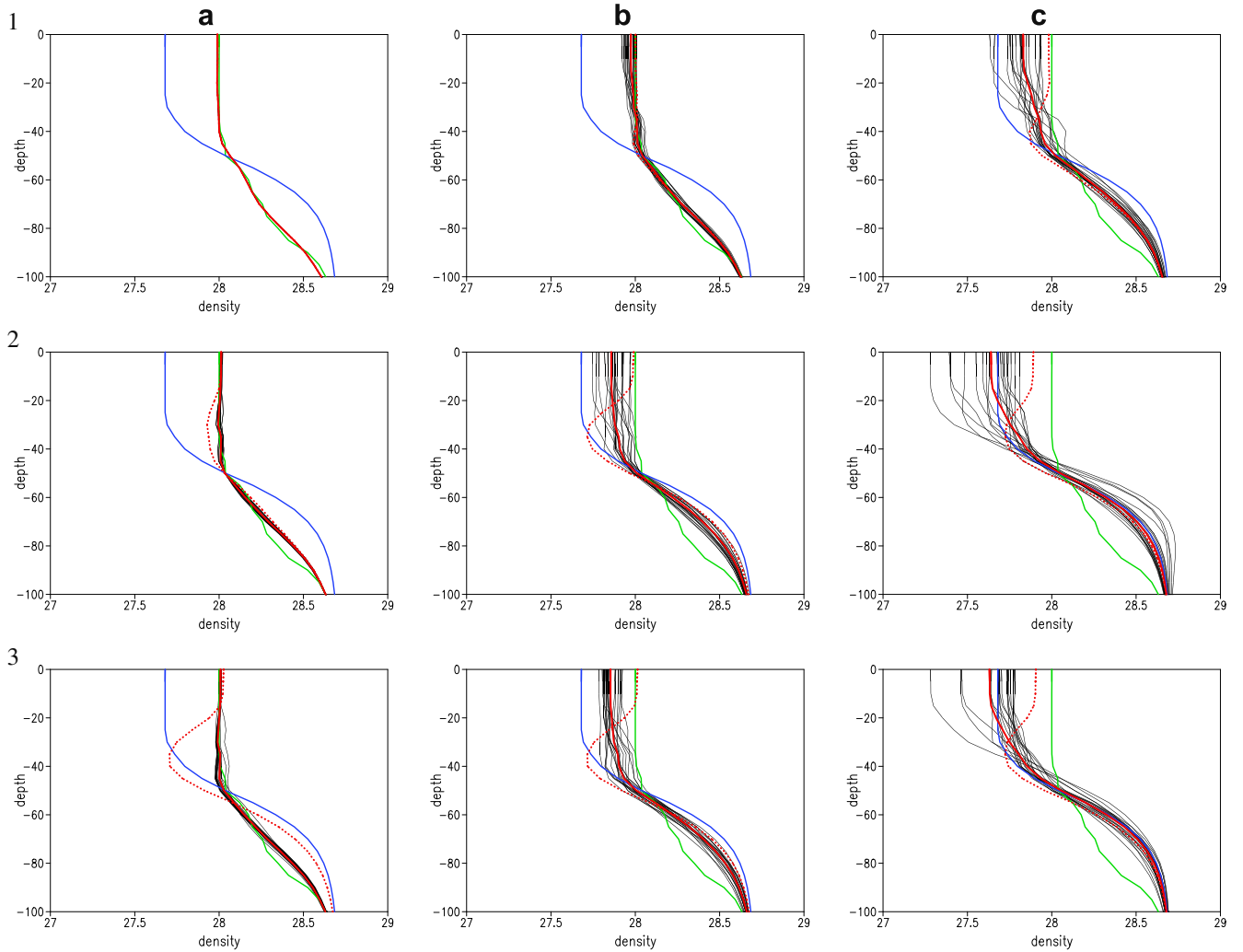


Fig. 11. Sample (30 members) of the truncated Gaussian $\mathcal{TN}(\mathbf{x}^u, \mathbf{P}^u; I)$ for 9 different observation systems and $r = 7$. The figure also shows the forecast state $\mathbf{x}^f$ (blue line), the true state $\mathbf{x}^t$ (green line), the truncated Gaussian estimator $\mathbf{x}^a$ (red line), and the location vector of the truncated Gaussian pdf $\mathbf{x}^u$ (dotted red line).

with a very good level of accuracy). Fig. 12 shows the same information as Figs. 10 and 11, but with $r = 3$ instead of $r = 7$. We stated before that observation system 1.a is so accurate that both the Gaussian and truncated Gaussian estimators (obtained with $r = 7$) are indistinguishable from the truth. Hence, the only reason why the estimators obtained with $r = 3$ are so different from the truth is that the size $r = 3$ of the reduced space is too small to describe the subspace in which the probability is significant. Moreover, we observe that in this situation the Gaussian analysis can also produce unstable estimates. On the other hand, a consistent truncated Gaussian estimator was found in this subspace, satisfying the stability constraints, though it was not as close to the truth as it should be with such a perfect observation system.

5. Application to a hybrid coordinate model of the Bay of Biscay

After encouraging results obtained with the one-dimensional mixed layer model, the natural step forward is to test the TG filter with a real size system. In this section, it is applied to the problem described in Broquet et al. (2007). The model is a three-dimensional ocean circulation model of the Bay of Biscay at a $1/15^\circ$ horizontal resolution (214×176 grid points, including 33,815 wet points). It is based on the hybrid coordinate ocean model HYCOM (Bleck, 2002; Chassignet et al., 2003) and covers an area (North-

West Europe) that includes the abyssal plain of the Bay of Biscay, the Armorican shelf and the southern part of the Celtic shelf (see Fig. 14). In their paper, twin assimilation experiments are performed to test the possibility of controlling errors due to imperfect atmospheric forcing using sea surface temperature (SST) observations. Their data assimilation method was based on a reduced-order Kalman filter (SEEK filter) that we want to upgrade now to a reduced-order TG filter (with the quasi-normal approximation, see Section 2.6.1). All other statistical parameterizations are kept unchanged with respect to the study by Broquet et al. (2007). In particular, the reduced dimension error subspace (ruling the horizontal influence of the observations) is defined by the first EOFs of the model variability, and a local algorithm is used to eliminate the spurious long range influences.

Concerning the expression of the hydrostatic stability inequality constraints, there is an essential difference between HYCOM and the mixed layer model used in the previous section. HYCOM is a hybrid vertical coordinate model: it can be viewed as a stack of layers (26 in our configuration) that smoothly transform from isopycnal layers (i.e. of constant potential density) in the open stratified ocean to z -coordinate layers in the mixed layer and unstratified areas, and to σ -coordinate layers in shallow coastal regions (see Chassignet et al. (2003) for a description of the three vertical coordinate systems used in most ocean models). Thus, the

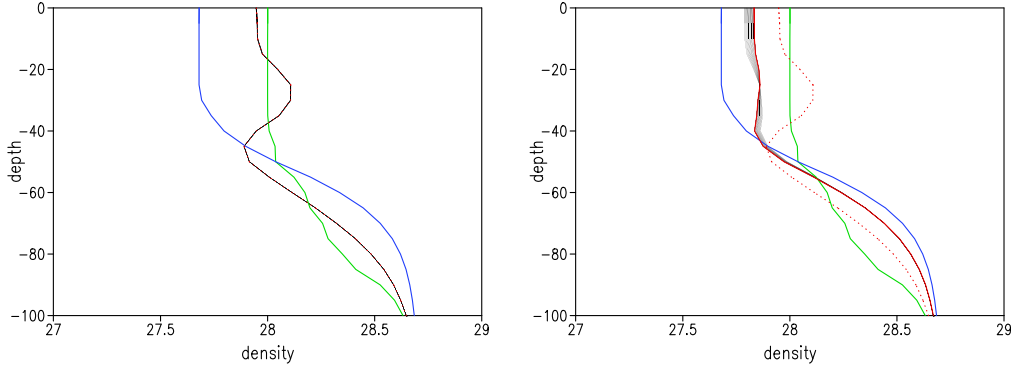


Fig. 12. Comparison of observational updates using the Gaussian assumption (left panel) and the truncated Gaussian assumption (right panel), as illustrated by samples (30 members) of the posterior distributions $\mathcal{N}(\mathbf{x}^a, \mathbf{P}^a)$ and $\mathcal{TN}(\mathbf{x}^a, \mathbf{P}^a; I)$ for $r = 3$, with observation system 1.a. The figure compares the truncated Gaussian variance minimizing estimator $\mathbf{x}^a$ (solid red line on the right panel) with the location vector of the truncated Gaussian pdf $\mathbf{x}^a$ (dotted red line on both panels), the forecast state $\mathbf{x}'$ (blue line) and the true state $\mathbf{x}^t$ (green line). Note that on the left panel, the 30 members of the Gaussian pdf are almost indistinguishable from the Gaussian estimator $\mathbf{x}^a$ because of the high quality of the observation system.

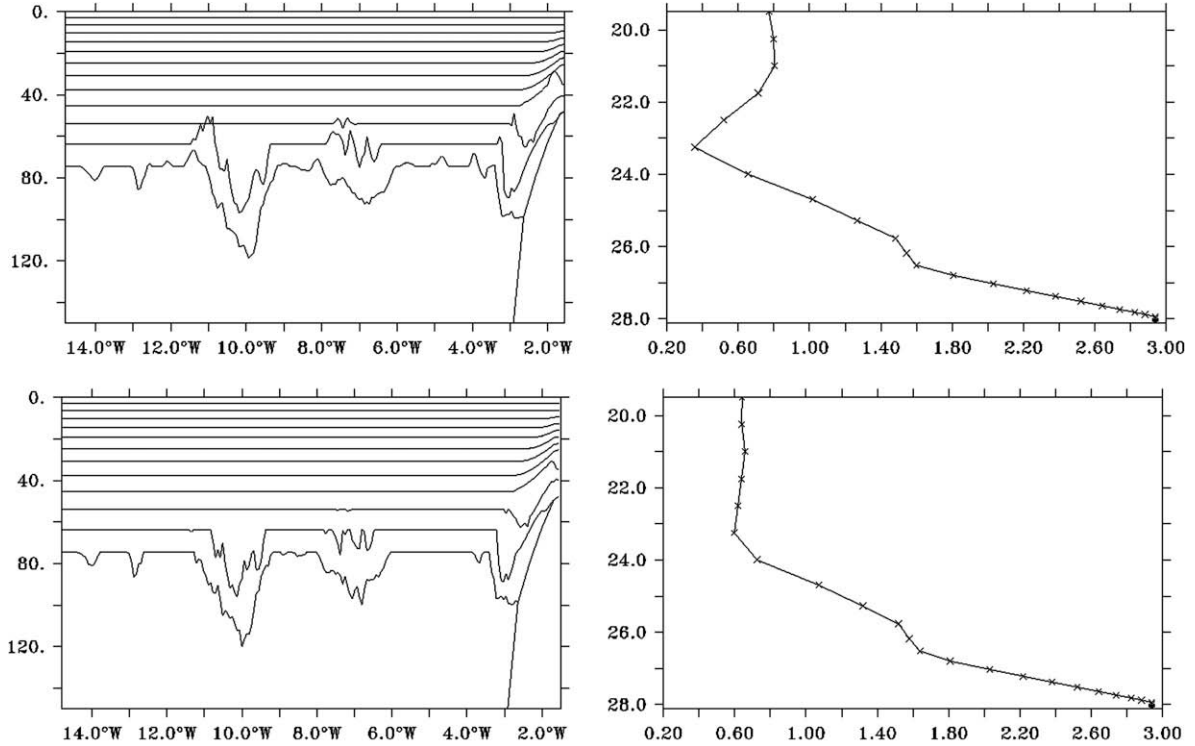


Fig. 13. Comparison between the classical Gaussian observational update (top panels) and the truncated Gaussian observational update (bottom panels) for a zonal section (at 45.9°N) of interface pressure (left panels) and one density profile (right panels) belonging to that section (at 6.4°W).

model state vector includes the layer interface pressure p in addition to the layer temperature and salinity to describe the ocean stratification. This modifies the constraint of hydrostatic stability as follows (compare to Eq. (20), which is valid for a z -coordinate model):

$$\rho_k - \rho_{k+1} \leq \epsilon \quad (24)$$

$$p_k - p_{k+1} \leq 0 \quad (25)$$

where k increases with depth. The second constraint imposes non-negative layer thicknesses. It is the only constraint to apply in pure isopycnic coordinate models since all layers have a prescribed potential density. (It can also be viewed as a condition of coherence of the vertical density coordinate, which can only be valid in a stratified ocean.) In HYCOM, there is also a small tolerance for unstable

profiles (in our configuration, $\epsilon = 0.02 \text{ kg/m}^3$) that has been introduced in the first constraint (Eq. (24)). Moreover, the management of the hybrid coordinate imposes a minimum pressure for each layer interface (corresponding to the z -coordinate levels in the upper ocean), that is a function of the bathymetry H in the shallow region (σ -coordinate levels on the continental shelf):

$$p_k \geq p_k^{\min}(H) \quad (26)$$

Fig. 13 (top panels) shows the result of a classical Gaussian observational update for a zonal section (at 45.9°N) of interface pressure (left panel) and one density profile (right panel) belonging to that section (at 6.4°W). We can see that all three constraints (24)–(26) are violated: density inversions, negative layer thicknesses, and interface pressure below minimum value (represented in the figure

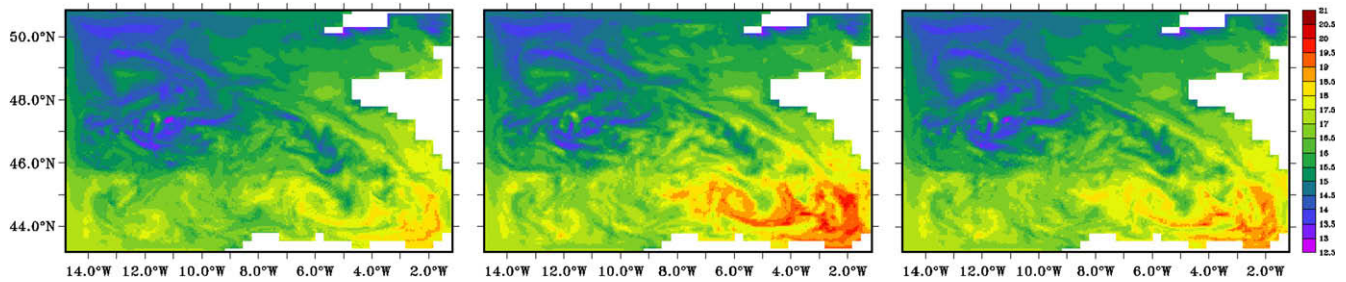


Fig. 14. Estimated SST as obtained using the various algorithms described in the text: Gaussian observational update (left panel), the same followed by the adjustment operator (central panel), and truncated Gaussian observational update (right panel).

by the straight z -coordinates horizontal lines, curving to σ -coordinates on the continental shelf). As observed in Section 4.2, this is linked to the weakness of the observation system: only SST is observed, with 0.1 °C error standard deviation. In Fig. 13 (bottom panels), we show the result of the truncated Gaussian observational update (minimum variance estimator $\bar{\mathbf{x}}^a$): density inversions have disappeared (except for the ϵ tolerance introduced in the constraint), all layers have retrieved their minimum depth (even in the σ -coordinate region), with non-negative thicknesses. It must be stressed that the truncated Gaussian estimator is computed locally on the horizontal i.e. separately for every water column of the system (using only the part of the truncated Gaussian pdf corresponding to that water column). This approximation can be used only because all inequality constraints (Eqs. (24)–(26)) are local: each of them involves variables belonging to one single water column. (With such a system, a global truncated Gaussian observational update would be impossible because the number of constraints to deal with simultaneously $3 \times 26 \times 33,815 \approx 2,600,000$ is too large to be numerically tractable.)

It should be pointed out that in all previous data assimilation studies using HYCOM, posterior empirical corrections were of course imposed after the observational update to restore the constraints before restarting the model: an adjustment operator was applied (as explained in Brankart et al., 2003). In Broquet et al. (2007), it consisted of three successive operations: (i) iterate from bottom to top layer and decrease the density of each layer (using the same temperature/salinity ratio as in the observational update) until it is smaller than the density of the layer below (with tolerance ϵ); (ii) iterate from top to bottom layer and increase the thickness of each layer (while decreasing the thickness of the layer below by the same value) until it is non-negative, and until the deep interface of the layer is larger than or equal to the minimum value; (iii) iterate from bottom to top layer and increase the thickness of each layer (increasing the thickness of the layer above by the same value) until it is non-negative. This algorithm can be shown to be sufficient to restore constraints (24)–(26).

Fig. 14 shows the effect of this algorithm on SST: the removal of a density inversion from bottom to top layer (first step of the algorithm) requires warming up the top layers so that the SST values are driven away from the analysis (which is closed to the observation). The SST RMS error increases from 0.8 to 1.23 °C during the adjustment stage. Note that starting the algorithm from the top layer would not alter the SST but would in general have a worse impact by destroying the deep ocean thermohaline properties. On the other hand, the figure (left panel) also shows that the truncated Gaussian observational update manages to satisfy all constraints, with SST values remaining close to the observations (but larger than the Gaussian estimate at the south-east corner of the Bay, even if more moderate than the adjustment operator). With the truncated Gaussian observational update, the SST RMS error remains as low as 0.98 °C. This spurious warming occurs above the first density inversion of the

profile and usually affects the whole mixed layer (as with the density profile in Fig. 13). As a result, for the full 3D temperature field the RMS error is 0.72 °C for the Gaussian estimate, 0.83 °C after the adjustment stage, and 0.74 °C for the truncated Gaussian estimate.

Of course, more sophisticated adjustment algorithms can be imagined (like vertical mixing or averaging for example), but the important point here is that, with the truncated Gaussian observational update (being fully multivariate), there is no need to find a clever adjustment algorithm, which should in any case be re-examined in any new situation. The optimal state vector is found automatically, taking into account all multivariate correlations or contingencies between all variables, as described by the prior truncated Gaussian pdf. All the little tricks that would be needed to build the adjustment operator are replaced by the single truncated Gaussian hypothesis.

6. Conclusion

In this paper, we have shown that it is possible to build an optimal nonlinear filter based on the assumption that error probability distributions are truncated Gaussian distributions. The justification for this assumption is that the resulting TG filter is numerically much more efficient than more general methods that do not make any assumption about the shape of the probability distributions (such as particle filters). It is especially more efficient because the observational update can be performed using the same formula as the Kalman filter (Eqs. (7)–(9)), and because truncated Gaussian distributions can be sampled efficiently (with a Gibbs sampler). Moreover, to be identified adequately, a truncated Gaussian distribution requires a *much smaller* sample than a general distribution, since the general shape and the inequality constraints are already known.

In terms of methods, a key point is often to make hypotheses that are generic enough to adapt to a wide variety of problems, but strong enough (or shaped in a special way) to make the computation of the solution numerically tractable. A useful framework is proposed in Sambridge and Mosegaard (2002) to compare inverse methods in terms of a trade-off between exploration of the problem structure and exploitation of prior assumptions. In this framework, the TG filter must be viewed as less explorative than the particle filter (no hypothesis on the shape of the probability distribution), and more explorative than the Kalman filter (assuming linearity).

Testing it as an application, we have shown that it is a sufficient approximation to deal with the constraint of hydrostatic stability in ocean analyses. We focused the study on the observational update which represents the real difference with respect to other filters, like the ensemble Kalman filter or the particle filter. A detailed evaluation of the method was conducted using a one-dimensional z -coordinate model of the ocean mixed layer. We have shown in

particular: (i) that the truncated Gaussian hypothesis is more appropriate than the classical Gaussian hypothesis for representing the error in the system (as simulated by perturbations of the wind and air temperature), (ii) that this hypothesis leads to estimates of the state of the system that are consistent with the hydrostatic stability inequality constraints, and (iii) that the method is especially useful if the observational information is sparse and inaccurate, or if the size of the reduced space is insufficient. The method was also applied to a more realistic three-dimensional hybrid coordinate model of the Bay of Biscay (at a $1/15^\circ$ resolution). For this particular example, it was found that: (i) the method can impose hydrostatic stability in isopycnic coordinates as well as in z -coordinates, (ii) the method is efficient enough to be applicable to real size problems. In both applications, it was observed that the hydrostatic stability constraints contained valuable information to be taken into account in the estimation process, often facilitating the vertical extrapolation of SST observations (which are among the most available observations in practical applications).

These positive results suggest that the truncated Gaussian assumption may be useful in many other ocean applications, or even in other fields of research. Ocean systems are subjected to many other inequality constraints: salinity within an acceptable range, sea-water temperature above melting point, sea-ice temperature below melting point, non-negative ice thickness, ice concentration between 0 and 1, non-negative biochemical concentrations... Model parameters are often constrained by bounds, so that the truncated Gaussian filter would also be useful as an extension of dual Kalman filters, estimating model parameters together with the state of the system. The method can also be easily generalized to nonlinear inequality constraints provided that they define a convex domain in the state space. The only thing to modify in the algorithm is the computation of the bounds for the one-dimensional conditional probability distributions (Eq. (13)) that are used in the Gibbs sampler (and which remain truncated Gaussian distributions). More serious difficulties occur if the domain defined by the constraints is not convex, because the probability distribution can become multimodal and because the expected value (Eq. (3)) would no longer be guaranteed to satisfy the constraints.

Acknowledgements

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Appendix. Computational complexity of the algorithm

The computational complexity of the TG filter can be estimated as a function of the size x of the state vector, the average size y of the observation vector (since the complexity is linear in y), the dimension r of the reduced space, the number c of inequality constraints, the number m of operations needed to perform one model integration, the size n of the sample needed to accurately sample the truncated Gaussian distribution, the number p of projections required to reach a state vector satisfying all constraints. The overall complexity is polynomial: it is the sum of products of various powers of these quantities. In the following evaluation, we give only the dominant powers (assuming that x, y, r and n tend to infin-

ity), without making any hypotheses about the relative magnitude of these numbers.

Complexity of the analysis (Eqs. (15)–(18)): $N^a = (x + y + \beta r)r^2$ with $\beta = O(1)$.

Complexity of the computation of the estimator
 $\bar{x} : N^k = (x + n + 2p)cr$.

We cannot provide the complexity of the computation of the estimator $\bar{x}$ since it depends on the algorithm that is used to solve the quadratic programming problem. It is very likely that under normal circumstances $N^k \ll N^x$.

Complexity of the sampling of the truncated Gaussian pdf:
 $N^s = (n + 2p)cr$.

Complexity of the ensemble of n forecasts: $N^f(n) = nm$.

Complexity of the estimation of $\bar{x}$ and $\bar{C}$ from a sample:

$$N^e = xn(n + r) + \beta n^3 + (n' + 2p + x)cr + l(n + n')r^2 \quad \text{with} \\ \beta = O(1)$$

This complexity is more difficult to evaluate since it depends on the number l of evaluation of $\log L$ needed to reach the minimum, and on the size n' of the sample that is needed to obtain an accurate evaluation of α . If this cost is too high for large scale problems, it is advisable to use the suboptimality described in Section 2.6.1 and use $\bar{x}$ and $\bar{C}$ as proxies for $\bar{x}$ and $\bar{C}$.

From these elementary pieces, it is a straightforward step to obtain the computational complexity of the various suboptimal variants of the scheme as compared to the full filter (where N^k can be substituted by N^x if $\bar{x}$ is computed instead of $\bar{x}$):

Full reduced-order TG filter: $N = N^a + N^s + N^f(n) + N^e + N^k$

→ **quasi-normal approximation:** $N = N^a + N^s + N^f(n) + N^x$

→ **no forecast of the truncated Gaussian pdf:**

$$N = N^a + N^f(1) + N^x$$

Full reduced-order Kalman filter: $N = N^a + N^f(n)$

→ **no forecast of the covariance matrix:** $N = N^a + N^f(1)$

The additional complexity with respect to Kalman filtering is equal to $N^k + N^s + N^e$, which is reduced to N^x with quasi-normal approximation and no forecast of the truncated Gaussian pdf. For large systems, $x \gg y, x \gg r, x \gg n, x \gg p$, the relative cost with respect to the analysis is then $\frac{N^k}{N^a} \sim \frac{c}{r}$. In the mixed layer example (Section 4), on the contrary, the state vector ($x = 405$) is large with respect to the observation vector (between $y = 2$ and $y = 162$ according to the experiment) or with respect to the dimension of the reduced space ($r = 3$ or $r = 7$): $x \gg y, x \gg r$, but small with respect to the sample size: $n \gg x$, so that the relative additional cost is about $\frac{N^k}{N^e} \sim \frac{x+n}{x+y} (\frac{c}{r})$. (In our experiments with this small model, the sample size is chosen very large because the numerical cost is anyway not a problem.) In the HYCOM example (Section 5), the situation is also somewhat different, because a local algorithm is used for the observational update, so that the size x of the local state vector is not large with respect to y or n . In that kind of situation, the relative additional cost of the observational update is thus about $\frac{N^k}{N^e} \sim \frac{x+n}{x+y} (\frac{c}{r})$, which is about 500% in our application. Note, however, that a quite large sample size n has been used for a good relative accuracy, and that this relative value would quickly decrease with the size of the observation vector y .

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